

STUDIO TECHNOLOGY

NOTES

IPS FINE ART EXAMINABLE PAPERS.

PAPERS OFFERED

1. Paper 612/1. Studio technology theory. This is a compulsory paper. -2hours
2. Paper 612/2. It's a combination of still life and nature study. 2 ½ hours
3. Paper 612/3. Study of the human figure. 3hours
4. Paper 612/4. Original imaginative composition in colour. 5hours
5. Paper 612/5. Graphic design Craft A 5 ¼ hours

NOTE

- **Do all the given activities thoroughly according to the given topics.**

INTERGRATED PRODUCTION SKILLS (IPS) FINE ART

Art is a word that is often used every day. Studio technology for secondary school deals with materials, tool exploration and experimentation to develop skills. It enables students to learn from a variety of experiences.

Therefore, **studio technology** refers to the preparation and experimentation of different materials and tools used in art and design.

Fine art can be defined in various ways and some of them are;

- Art is the act of expressing one's feelings /ideas.
- Art is also a visual language of expression.
- Art refers to the visual means of conveying a message.
- Art is also the human skill of creating beauty through painting, designing etc.

They are reasons or relevancies as why we study art and craft. Some of these include;

1. To preserve culture and tradition.
2. Source of employment and income.
3. Helps students develop their creative power/ talents.
4. It helps in coordinating other subjects for example biology, geography etc.
5. It's a strong medium of expression and passing of information.
6. Since art involves group work. It teaches students ideas,
7. It trains students to be resourceful to the environment which they can use e.g. raw materials like papyrus, fiber, sisal, grass, clay etc.
8. To pass exams and have further studies.

Art is grouped into types or categories, forms, disciplines and units or branches. The types of art these include;

TYPES OF ART

- **VISUAL ART:** This is the type of art that is perceived visually. E.g. drawings, paintings, crafts.
- **PERFORMING ART:** This is the type of art that deals with action and movement of body parts to convey artistic expressions through Music Dance and Drama (MDD)

- **LIBERAL ART:** this is a form of art which is represented in form of reading, writing and debating or making analysis e.g. poetry, literature, philosophy, history etc.

FORMS OF ART

The forms of art include;

- ✓ **Three Dimensional (3D)**
- ✓ **Two Dimensional (2D)**

These differ in the following ways;

Three dimensional (3D)		Two dimensional (2D)
1.	These are tangible, they can be lifted and turned around	These are made on a flat surface like paper.
2.	These are measured and observed in height, width and length.	These are measured and observed in only length and width.
3.	These include sculpture, ceramics, weaving, multimedia, architecture	These include paintings, photography, drawing etc.
4	They are defined by volume and mass.	These are defined by their medium and composition.

UNITS/BRANCHES IN ART

Fine art is divided into the following branches or units namely;

1. **Drawing and painting from still life.** This is the art drawing and painting a composition from manmade objects/ artificial objects e.g. tables, chairs, utensils.
2. **Nature study.** This is the art of drawing and painting compositions from objects that exist naturally e.g. animals, landscape, plants, trees, mountains etc.

3. **Study of a Living person/ human figure.** This involves the study of a human body or human anatomy.
4. **Imaginative composition.** This is imagining of a composition from a given topic and painting it in colour. The composition can go from past experiences, fiction, present situations etc.
5. **Graphics.** This is the art of arranging visual images and words to communicate a message. Or it's the designing of letters with illustrations. Examples of graphic art include; book covers, posters, magazines, labels etc.
6. **Studio technology.** This involves the making of crafts e.g. pots, mats, sculpture, baskets etc.
7. **History art and appreciation.** This is the study of how art and design started from the prehistoric era to the present time.

DISCIPLINES/ FIELDS OF ART AND CRAFTS

These disciplines of art and design include;

- Drawing
- Painting
- Sculpture
- Weaving
- Graphics
- Fabric decoration
- Ceramics
- Multimedia
- Basketry
- Leather work

NOTE: as student, you need to be able to explain each of the above individually and in detail.

Any art or craft making is carried out in a working place called a “**STUDIO**”. So a studio is defined as the working place of an artist; when it's in a school setting, a studio is termed to as an “**ART ROOM**”.

Once the crafts or any other artwork is completed, its displayed for either viewing or for sale. These are displayed in a place called a “**GALLERY**”.

Just like at home, an art room, gallery or studio should have rules that are followed when using it.

ACTIVITY ONE: Research and write down the rules that govern any of the above places as used in Art.

ACTIVITY ONE: Research and write down the importances of an art room to a student.

CHARACTERISTICS OF A GOOD ART ROOM

These include;

- It has to be regularly clean.
- Furniture must be kept in order to avoid congestion.
- It should have a place where finished art works are properly stored and displayed.
- It should have a proper storage for material used for art.
- Have enough space to work from.
- Must have a source of water for cleaning or preparing materials.

The artworks made in the studio or art room are called crafts.

So the term a “**CRAFT**” is defined as any artwork skillfully produced by hands in a traditional way.

TYPES OF CRAFT

The types of crafts include;

- Decorative crafts e.g. collage, mosaics, carvings etc.
- Fashion crafts e.g. necklaces, anklets, armlets etc.
- Textile crafts e.g. wall hangings, fabrics from appliqué etc.
- Functional crafts e.g. mats, pottery, basketry etc.

A person that makes or produces a craft is termed to as “**craftsman**”.

ACTIVITY THREE: Research and define terms a craftsman and craftsmanship.

ACTIVITY FOUR: Research and write down the importances of studying craft making?

Like any other job, the growth of the craft making industry is influenced and hindered. The following are some of the bottlenecks that hinder this industry;

- Local materials used to make crafts are hard to get.

- Industrial made materials are expensive.
- Few people appreciate crafts.
- Creates conflict with religion, taboos and culture.
- Lack of knowledge of application of craft making.
- Competition with modern art.
- Lack of enough skills needed to make the crafts.

MATERIALS AND TOOLS USED IN ART AND CRAFT

Art works plus the crafts are made of materials and the craftsman or the artist uses tools for making these products.

- A **material** is defined as a substance used to form/create an art work.
- A **tool** is defined as an object, device or instrument designed to perform a specific kind of work but it's not part of the final work.

IMPORTANCE OF MATERIALS AND TOOLS IN ART AND CRAFT

The various materials and tools used in art and craft have vast importances. Some of these include

- They act as a medium of expression
- Help in the development of the learner's creativity.
- Tools and materials aid in the execution of different art works since they are essential.
- Tools and materials help learners to clearly express themselves through various specialization.
- They act as a medium of expression.

ACTIVITY FIVE: *Research and explain the advantages and disadvantages of using local materials for art and craft.*

STEPS TAKEN WHEN MAKING ANY CRAFT WORK

Several factors can be considered while making a craft. These include;

- **Message;**
An artist has to have the reason as why they are making the craft work and which message are trying to communicate.
- **Creativity;**

This is the ability for someone to come up with something new. This involves the use of a unique design process.

- **Composition;**

This is the ability to put together, arrange and organize the elements of art following the principles. Once the sketches have been developed up to the final work, they have to formulate detailed compositions for the final works.

- **Structure;**

This is judged by looking at how different forms stand out to define the inner mass of the entire craft, showing the relationship between different parts.

- **Use of materials;**

Use of materials is judged by the way one uses a material to create a unique piece of art work.

- **Finish;** once the craft has been created, it must be completed and finished to serve the purpose it was designed for.

ELEMENTS AND PRINCIPLES OF ART AND DESIGN

The art works that are produced both three and two dimensional, are designed in line with the elements and principles of design. These two aspects move hand in hand.

ELEMENTS OF ART

These are defined as building blocks / tools/ingredients of visual arts which help to analyze the success or failure of a particular work of art.

IMPORTANCE OF ELEMENTS OF ART

The following are the importances of elements of art.

- They describe the characteristics of a given work of art
- They help analyze the success or failure of a particular art work

- They help us speak a common language as we communicate through art.
- They help in the production of an art work.
- They are the foundation tools on which all Art is built without which Art is nonexistent.
-

Elements of art include;

➤ LINE:

A line is a continuous mark made on a surface to join one or more points. Or a mark made by a moving tool when its point of contact is made to move.

TYPES OF LINE

- | | |
|----------------------------|--------------------------|
| a) Horizontal lines | d) Circular lines |
| b) Vertical lines | e) Zigzag lines |
| c) Diagonal lines | |

Lines vary in character; they may be continuous, broken, thick, bold, faint, long, short, parallel, straight and curved.

USES OF LINE

- Lines enclose shapes,
- Define forms
- suggest movement,
- Create depth
- Create different moods and emotions.
- Show the effect of light and shade
- Indicate the weight of a given object
- Create texture
- Reveals areas of emphasis in a design
- Organize space by separating or dividing
- Used to connect different objects and ideas.

➤ SPACE:

This is the area within or around a given work of art.

Space can be negative or positive. Negative space is the area around and between the subjects of an images. Positive space that's is covered by the artwork or object.

➤ **SHAPE:**

This is an enclosed area with well-defined boundaries or the outline area of an object.

TYPES OF SHAPES

- **Organic shapes/Irregular/ free-form shapes**

These are shapes that are derived from organic objects such as plants, rocks, animals, clouds etc. these shapes are found to occur naturally in the environment and they are unpredictable and flowing in appearance.

- **Geometric shapes/ Regular shapes.**

These are shapes with regular dimensions which can be measured and drawn in different scales e.g. circles, squares, rectangles, triangles, semi-circles e.t.c.

➤ **FORM**

This is the roundness of a given object in three dimensional space, mass and volume. Form in two dimensional art creates a three-dimension aspect of height, width and depth.

➤ **COLOUR**

It is an illusion of sight that is created when light falls on an object.

PROPERTIES OF COLOUR

Colour has three main properties namely;

- Hue
- Value
- Intensity

- i. **Hue.**

This is the property of color that makes it differ from another.

ii. Tone/ Value.

It refers to the lightness and darkness of a colour

iii. Intensity (Chroma)

This is the purity and saturation of colour. This intensity can be altered by adding black or white to form a shade or a tint.

CATEGORIES/ SCHEMES OF COLOUR

Colour wheel:

A colour wheel is a basic aid that has different arrangement of colours to help an artist. The different categories of colour include;

a) Primary colours

These are colours that cannot be obtained by mixing other colours i.e. red, blue and yellow (RYB)

b) Secondary colours.

These are colors produced by mixing two primary colors i.e.

BLUE + RED = PURPLE/VIOLET

BLUE + YELLOW = GREEN

RED + YELLOW = ORANGE

Therefore, PURPLE, GREEN and ORANGE are secondary colors.

c) Tertiary colors.

These are colours produced when a primary colour is mixed with a secondary colour.

d) Neutral colors

These are colours that change/ enhance the tone/ value of other colours, and these are Black and WHITE.

e) Complimentary colours.

These are colours that lie directly opposite each other on the colour wheel e.g. blue is opposite orange and therefore blue and orange are complementary colours

f) Cool colors.

These are associated with things like water, vegetation, space, distance and the sky. Cool colors can be used to create feelings of coolness, sadness, calmness and despair e.g. blue, green, purple.

g) Warm colours.

These are usually associated with sources of heat and light such as fire and sun. Warm colours are used in compositions to create feelings of warmth, anger, activity and life e.g. red, orange, yellow and adjacent colours on the wheel

USES OF COLOUR

They are used to do the following;

- Describe form.
- Modal form and create illusion of depth.
- Suggest movement.
- Express ideas and feeling i.e create mood.
- Arouse emotions.

ACTIVITY SIX: *Draw a colour wheel and on it show Primary, Tertiary, secondary, Complementary colours.*

➤ **TEXTURE:**

This is the roughness and smoothness of a surface. Or it's the surface quality of an object.

Types of texture

- Tactile texture: this is the type of texture experienced by touch
- Visual texture: this is the type of texture only experienced by sight e.g a drawing of a stone, the texture can be interpreted as rough by our eyes but the paper remains smooth when you touch it.

ACTIVITY SEVEN: *Research and explain other types of textures that are used in art and design.*

USES OF TEXTURE

- It can be used to define shape,

- highlight form,
- create the illusion of space and depth,
- make a surface appear lively, interesting or dull,
- Create visual harmony with other elements around.
- Texture creates visual reality and identification of objects

➤ **VALUE:**

Value is the degree of lightness or darkness of a surface. It can be created on a surface by two techniques

➤ **TONE:**

Tone is the lightness or darkness of an object due to the effect of light. Tones are got by manipulating a given medium to show the dark and light parts of an object achieved by controlling the value.

ACTIVITY EIGHT: *Research and come up with the importance of tone and value as elements of art.*

One of the ways how tone and value can be achieved is through tinting and shading.

TINTING:

This is the process of lightening a surface to increase the ability of the surface to absorb light. By tinting the value of a colour is raised and this is achieved by adding white.

ACTIVITY NINE: *Research and define the following terms;*

- | | |
|--------------------|-----------|
| a. Tonal variation | d. Tint |
| b. Tonal scale | e. Toning |
| c. Shade | |

SHADING:

This is the process of darkening a surface. By shading the value of a colour is lowered and this is achieved by adding black.

TYPES OF SHADING

The different shading techniques include;

- **Smudging/Blending:**

It is achieved by rubbing pencil strokes to one another while varying pressure to obtain darkest and lightest tones.

- **Cross hatching:**

This is closely related to hatching. Here, we have a mixture of lines going from different directions.

- **Hatching:**

This is where closely related lines are drawn in one direction to create tone. The pressure inserted on both the paper and the drawing tool varies in order to create the different tones.

- **Scribbling/ doodling:**

It is mostly achieved by moving a drawing tool in a circular direction to create a tangle of lines that formulate various tones.

- **Stippling:**

In this method, dots are created /used for shading/toning.

ACTIVITY TEN: *Use the above explanations of the types of shading and illustrate each one of them.*

PRINCIPLES OF ART AND DESIGN

These are the guidelines used to arrange and organize the elements of art in order to produce a satisfactory piece of art work.

IMPORTANCE OF PRINCIPLES OF ART AND DESIGN

These include;

- They help to organize the elements of art and design during the process of making an artwork.
- They aid artists when they are planning for their art projects.
- They help to create a way for reactions and criticisms from the viewers.
- They also add to the technical language used in art and design.
- They also help an artist when they are making an art work.

Examples of principles of art and design are;

1) **BALANCE:**

This is the state of equilibrium between elements used in a composition.

Balance has been categorized into;

i. **Symmetrical/formal/passive balance.**

This is a type of balance that is achieved when the opposite parts of a given art work are exactly or nearly the same with other part. This particular type creates a feeling of stability and uniformity.

ii. **Asymmetrical/informal/active.**

This is a type of balance that does not weigh equally on both sides of an art work i.e. balanced by elements which are not identical.

iii. **Radial balance.**

This is a type of balance where the elements are equally distributed from the centre to the outer area; the centre becomes a potential focal point. Radial balance can either be symmetrical or asymmetrical.

ACTIVITY ELEVEN: *Use the above explanations of the types of balance and draw the relevant illustrations.*

ACTIVITY TWELVE: *Use the above explanations of the types of balance and come with the differences amongst them.*

2) **RHYTHM:**

This refers to the visual movement in an art work or a flowing movement having a regularly repeated pattern.

How to achieve rhythm

i. **Repetition.**

Rhythm can be achieved by repetition of similar elements which are placed together at regular intervals throughout the composition.

ii. Alternation.

This is created by arranging elements to follow each other alternatively e.g. an angular shape and circular shape may be alternated both in shape and size.

iii. Progression.

This is achieved when the flow of elements gradually changes in a sequence. Forms appear in an orderly sequence which creates rhythm and movement.

3) EMPHASIS:

This is the making of an area in an art work to stand out of the rest of the areas in an artwork.

4) DORMINANCE:

It's a principle of art and it refers to attractiveness, boldness of different forms or elements of art to attract the viewer.

5) Proportion.

It's also a feeling of unity created when all parts relate well to each other in an art work.

6) Harmony.

This is the visual satisfying effect of combining similar elements of art. It can be achieved through repetition and rhythm.

7) Unity.

This is the quality of a composition that makes it appear as a single unit.

8) Contrast.

This is the juxtaposition of different elements in a composition. It creates excitement and interest, and often draws the eye to certain areas in an artwork.

9. Variety:

This also a principle of art that deals with the use of several elements to guide the viewer's eye into the art work.

10. **Perspective.**

This is a principle of art that deals with the illusion of distance that makes distant object appear smaller and near objects appear bigger.

HOW TO CREATE PERSPECTIVE

a) By Overlapping

This is when one object close to the viewer covers part of a nearby or second object. **(draw objects overlapping each other)**

b) By Size variation.

Near large objects are made to seem closer to the viewer than smaller objects even though there on the same level.

(draw objects varying in sizes)

c) By Placement.

Objects in the foreground seem to be closer to the viewer than those in the background. **(draw and place objects in the foreground, middle and background)**

d) By Colour and Tonal variation

Colours in the foreground or closer to the viewer are darker than those at a distance or in the background.

e) By Detail.

Objects with clear and strong detail seem to be closer than those without clear details. **(draw the relevant illustrations)**

TYPES OF PERSPECTIVE

1. **LINEAR PERSPECTIVE:** It's a type of perspective viewed from the side or a flat surface.

Types of linear perspective

Linear perspective is both one point and two-point perspective.

a) One-point perspective

here the perspective has one vanishing point on the horizontal line. **(draw the relevant illustrations)**

b) Two-point perspective.

It's a type of linear perspective with two vanishing points.

(draw the relevant illustrations)

c) Three-point perspective

This is a type of one-point perspective with three vanishing points.

2. AERIAL PERSPECTIVE:

This is the types of perspective as viewed from above or below the eye level.

DRAWING AND PAINTING

Drawing is the process of making images by scratching, etching or by living a trace of lines on surfaces.

Drawing refers to the process of making marks on a surface to produce visual images.

Materials/ media in drawing

Media refers to all the tools and materials used to come up with any artwork.

A tool is an **instrument** or **device** held in the hand and used to produce a work of art and it's not part of the final work.

A material is a **substance** used as a media for drawing or as a surface on which a drawing is carried out.

With **dry media**, the artist doesn't use any liquid while with the **wet media**, the artist uses liquid to mix or to execute the artwork.

The following tools and materials can be used for the drawing:

Dry media

- Crayons
- Pencils
- Charcoal
- Oil pastels
- Coloured pencils
- Chalk

Wet media

- Water colours
- Paint
- Ink
- Powder colours

Supports in drawings

These are surfaces where drawing takes place.

- Papers
- Walls
- Wooden panels
- Ply wood
- Canvas

STAGES IN DRAWING

There are various stages in drawing which may or may not lead to the major composition, and these include;

- **Sketch**, this is the preliminary quick drawing done to capture something that is of interest to the artists.
- **Drawing as a Study**, this is the process which involves observation of a specified part of the object and how each relates to another.
- **Drawing as a complete work**: here, an artist carries out a drawing not as a sketch but as a complete work. A portrait drawing is an example of a complete work.

TYPES OF DRAWING

- **Memory drawing**: this is the type of drawing which has been expressed in the past and retained in the memory of an artist.
- **Imagination**: This refers to the art of forming mental images of what has never been seen. Drawing from imagination may be similar to that of memory because you may create new images or ideas by combining previous experience.
- **Observation**: this is the kind of drawing where an artist closely looks at an object and observes its particular characteristics and qualities. Drawing from observation develops the students' awareness of anything they come across in the environment. It also enables us to develop the awareness of the surface quality of objects like the lines, texture and tonal values brought about by light.

CATEGORIES OF OBSERVATION DRAWING

- **Still life**: This is the art drawing and painting a composition from manmade objects/ artificial objects e.g. tables, chairs, utensils
- **Human figure**: This is the study of the human anatomy.
- **Nature**: This is the art of drawing and painting compositions from objects that exist naturally e.g. animals, landscape, plants, trees, mountains etc.

- **Landscape:** This is the scenery that one sees when he/she looks at a particular section of the land or environment.

The landscape has three parts namely;

- **Foreground:** This is the area nearer to the viewer.
- **Middle ground:** This is the area just behind the foreground which stretches to the horizon.
- **Background:** This is the area furthest from the viewer and it's usually behind the horizon.

ACTIVITY 13: Use the above explanations about landscape and draw its parts.

SKETCHING

A sketch is a quick rough drawing or painting often made to assist in making of the finished work.

IMPORTANCE OF SKETCHING

- Sketching helps one to obtain the basic layout or appearance of the desired work.
- It helps in keeping records of work done and those in plan.
- Sketching helps in thinking and generating ideas necessary for the creation of an art work.
- It helps to improve the level of skill and speed at which art works are made.
- It helps one build confidence before starting off on the final work.
- Sketching is a means of expression especially during leisure time.
- Artists use their sketches to present their ideas to potential clients

SHADING

SHADING TECHNIQUES

1. **Hatches/ line shading:** it's a technique of shading objects by using lines that run in the same direction.

2. **Cross hatching:** This is a technique of shading objects by using lines that run from different directions crossing each other.
3. **Dots/stippling:** It's a technique of drawing and shading objects by use of dots.
4. **Scribbling (swirls):** This a technique of shading objects by using doddles or swirls. This technique is used when shading human hair, clouds and tree canopies.
5. **Blending/ smudging:** This is a technique of shading objects by using a very smooth tone.

ACTIVITY 13: Use the above explanations and work out the activity below;

Technique	Activity involved	Demonstration
Hatches/ line shading		
Cross hatching		
Dots/stippling		
Scribbling (swirls)		

Blending/ smudging		
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PAINTING

This is the art of using colours to describe form, express ideas, feelings and mood on different surfaces like canvas, walls etc.

TYPES OF PAINT

There are various types of paints used when painting and these include;

- **Gouache colours/Designer colours:** these are thick water based colours found in tubes or bottles.
- **Tempera colours:** these are mainly found in powder form. They tend to be opaque.
- **Water colours:** these are available in tablet and tubes in pigment form.
- **Oil bar or oil stick:** this is an art medium that is made of oil paint in solid form.
- **Poster colours:** these are water based and usually in bottles and already mixed. They are always used for graphic design.
- **Oil colours:** these are blended with oil and derived from natural sources. They are expensive and take quite long to dry.
- **Acrylic colours:** these are synthetic and easier to use than oil colours. They can be mixed with water and they dry rapidly.

ACTIVITY 14: Use the above explanations and formulate the importances of drawing and painting to an artist, community and the country.

MOSAIC

Mosaic is the art of creating images with assembling one type of small pieces of material such as coloured glass, stones or other materials on a surface.

Materials used in a mosaic are called **TESSERAE** and these only vary in size and colour.

The spaces found in between the tesserae are called **INTERSTICES**.

TYPES OF MOSAIC

The type of mosaic depends on the tesserae used to produce the mosaic e.g

- Paper mosaic
- Stone mosaic
- Glass mosaic
- Banana fiber mosaic
- Leather mosaic etc.

MATERIALS USED FOR MAKING MOSAIC

Mosaic offers an unlimited exploration of different types of materials available in the environment. However, the basic materials needed include;

- **Supports:** a good support in mosaic depends on the type of mosaic to be produced; in paper mosaic a stiff card, mounted canvas, plywood can be used. In stone and glass mosaic, cement or a plaster base is needed.
- **Glue** as an adhesive.
- **Tesserae** i.e. coloured glass, stone, banana fiber, paper, beads, seeds etc.
- **Vanish** for creating a good finishing on the artwork.

TOOLS USED IN MAKING MOSAIC

- Hammer
- Nails
- Glass cutter
- Knifes
- Pair of scissors

METHODS USED IN MOSAIC

They are mostly three methods used in creating mosaic artworks and these include;

- Direct method
- Indirect method
- Double indirect method

DIRECT METHOD

This method of assembling mosaic work involves directly placing (gluing) individual tesserae into the supporting surface.

Advantages of direct method

- This method is well suited to surface that have three dimensional qualities such as vases
- It suits small projects that are mobile and transportable (portable).
- The resulting mosaic is progressively visible allowing for any adjustments to the colour or placement.

Disadvantages of the direct method

- The artist must work directly on the chosen surface which is not often practical especially for long periods of time especially for large scale projects
- It is difficult to control the evenness of the finished surface.

INDIRECT METHOD

This method involves applying the tesserae facing down to a temporary surface e.g glue using an adhesive and later transferred to the final surface eg a wall surface or craft project with two-dimension surface, the temporary base material is then removed to reveal the mosaic topside up.

Advantages of indirect method

- This method is most use for extremely large projects.
- It gives the artist time to rework areas.
- Permits the cementing of tiles to the backing panel to be carried out quickly.
- In one operation it helps to ensure that the front surface of the mosaic tile and mosaic piece are flat and in the same plane on the front even when using tiles and pieces of different thickness.

ACTIVITY 15: *Use the above explanations and formulate the illustrations depicting the above method of mosaic making.*

DOUBLE INDIRECT METHOD

The tesserae are placed face up on a temporary support, when the mosaic is complete a similar medium is placed on top of it. The original underlying material then removed and the mosaic is then cast in its final base support.

Advantages of the double indirect method

- Large works can be done this way with mosaic being cut up for shipping and then re assembled for installation.

- This method enables the artist to work in comfort of his/her studio rather than at the site of installation.
- The operator directly controls the final results of the work which is important e.g. when the human is involved.

Disadvantages of the double indirect method

- It requires great skill on the part of the operator to avoid damaging the work .

ACTIVITY 16: *Use the above explanations and formulate the illustrations depicting the above method of mosaic making.*

ACTIVITY 17: *Use the above explanations and list the materials and tools that can be used to make a mosaic artwork.*

ACTIVITY18: *With the materials listed, come up with the strength and weakness of each material if used for mosaic making.*

ACTIVITY19: *With the tools listed, come up with illustrations of each tool as used for mosaic making.*

PROCEDURE OF MARKING A MOSAIC WORK

The following are the steps followed in making a mosaic work

- **Preparation**

This involves conceiving the ideas which can be physically or in the mind of the artist.

- **Gathering material**

Prepare the surface, glue, cutter materials to be used e.g. banana fibers, glass, stones

- **Sketching**

Make different sketches and design patterns on the surface to be worked upon using a pen, permanent marker, pencil, chalk etc.

- **Preparation of materials**

Cut the materials into small tiles (cubes) tesserae or tear according to different patterns and colour strength or marble to cause texture.

- **Executing the work**

Carefully add glue any other adhesive in use to a small piece (tesserae) and apply to the work surface following the patterns of the composition guided by the tones.

NB start with the dark areas by applying dark coloured tesserae and control the tones to light until when the composition is covered.

- **Finishing**

Finish the work i.e. depending on the materials used this may include varnishing, framing, reaming off excesses to give an impressive finish.

IMPORTANCIES OF THE MOSAIC ART WORK

- Mosaic serves a decorative artwork i.e. wall hanging, pavement ceiling's, floors etc.
- Mosaic work has an economic value when sold it's a source of income.
- Mosaic works are designed to embrace events.
- Mosaic murals are used as campaigning mediums.
- Mosaics are educational materials used as teaching aids and examination able crafts.
- Mosaic work improve and enhance the creative abilities of the artists.
- Mosaic picture reserve, promote and embrace cultural activities and beliefs.

CHALLENGES FACED WHEN PRODUCING A MOSAIC WORK

- Shortage in availability of materials especially traditional materials due to the changing environment and modernization.
- Expensive materials for example glass, marble stone slates, ceramic tiles etc.
- Limited knowledge and skill about mosaic production
- Very few people appreciate art and mosaic work which limits the market for mosaic article.
- Some mosaic materials are scarce there not easily accessible in some areas.
- Stiff completion from other pictorial crafts like printed photos, camera photos, drawings and paintings.
- Limited technologies of producing large quantities of mosaic work in shorter time

ACTIVITY 20: *With the above explanation, individually explain the procedures followed when making;*

- *Bean seed mosaic*
- *Banana fibre mosaic*
- *Stone mosaic*

COLLAGE

This is an art work produced by assembling/arranging different materials or text on a surface to form a composition.

The most important feature of collage is the exploration of the surface textures of materials and how they interplay with each other.

TYPES OF COLLAGE

Montage

It refers to the process of composing pictures out of miscellaneous elements such as magazines pictures, torn up into specific shapes and sizes pasted on a surface.

Photo montage

This involves the use of photographs to create a composition. The photographs can be cut up or used as a whole.

MATERIALS USED IN MAKING COLLAGE

Materials used in making collage artworks include;

- Supports
- Adhesives
- Textured materials such as tissue paper, newspaper, magazine paper, fabrics, fibers, wool, cotton, maize, bamboo, wood, saw dust, metals, sand, sea shells
- Vanish

TOOLS USED IN MAKING COLLAGE

- Hammer
- Nails
- Glass cutter
- Knives
- Pair of scissors

PROCEDURE FOLLOWED WHEN MAKING A COLLAGE

- I. Collect the different materials to be used
- II. Select a theme/motif and make sketches of the suggested idea on paper
- III. Transfer the sketch onto a hard support

- IV. Carefully paste materials on the support using a good adhesive following the composition sketches
- V. Varnish the collage to preserve different materials, frame it when complete.

Differences between a collage and a mosaic art work

- a) Collage places emphasis on the textural qualities of different materials while mosaic is made up of tesserae of one type of material.
- b) Materials in a collage overlap each other to produce paint like effects while a network of minute spaces running in different directions is produced in a mosaic.
- c) Mosaic is tone based i.e. tesserae are arranged to provide tonal variations in a picture composition whereas collage is texture based derived from the different materials used.

ACTIVITY 21: *List the importances of collage artwork to the economy of Uganda.*

CLAY AS A MATERIAL FOR POTTERY AND SCULPTURE

Clay is a fine grained, firm earthly material that is plastic when wet and hard when fired.

CLASSIFICATIONS OF CLAY

- a) **Residual/ primary clays.**
These are clays found near the original rock source; these clays are the purest type.
- b) **Sedimentary/ secondary clays.**
These are clays that have been carried from their original source by water erosions, these type of clays contain a lot of impurities

GENERAL PROPERTIES OF CLAY

i. Plasticity

This refers to the elasticity of clay. Clay must be plastic in order to be worked on in any form. This allows clay to be manipulated into all sorts of shapes and can be worked to a very fine degree.

ii. Porosity

This is the ability of clay to absorb water, air and other fluids. This porosity allows clay to dry without cracking; however, clay must be dried in a cool place to prevent rapid drying which can lead to cracking.

iii. Vitrification.

This is the ability for clay to convert into a glass state when being fired.

iv. Shrinkage.

This is the ability of clay to reduce in size by losing its water/ moisture to the surrounding during the process of drying.

TYPES OF CLAY

There are many different types of clay which contain different substances to give clay their particular properties. These types of clay are derived from the two classifications.

1. Kaolin (china clay)

This type of clay is nearly pure residual clay. It has low plasticity and a high degree of resistance to heat. It is used in the manufacture of ceramic tiles, toilet sinks and bath tabs. It fires to white in colour.

2. Ball clay.

This type of clay is very plastic and is used to blend with clays which have low plasticity. This is derived from secondary clay.

3. Shale clays.

This type of clay has low resistance to heat. It is the commonest type of the clays and it is used for making bricks, earthen pots etc. This is derived from secondary clay.

4. Earthen ware clay.

These are porous, non-water proof, opaque and fire below 1200⁰c.
This is derived from primary clay.

5. Stone ware clays.

These are sedimentary clays which fire above 1200⁰c. They vitrify and are able to hold water even when not glazed.

PROPERTIES OF PRIMARY CLAY

Primary clay the one that is found at the source of formation. It has the following properties;

- It has less plasticity
- It contains a lot of sand
- It fires at high temperature
- It's white in colour

PROPERTIES OF PRIMARY CLAY

Secondary clay the one that is carried away from its source of formation by agents of soil erosion. It has the following properties;

- Grey or dark brown in colour depending on the environment.
- It turns brown when fired.
- It develops pores when fired.
- They are carried away from the parent rock by forces of erosion.

METHODS OF CLAY PREPARATION

Freshly dug clay must undergo preparation before it can be used. Not all clays are the same some are more plastic than others while others can be used without adding anything. The methods used in clay preparation include;

DRY METHOD

This is the method where clay is pounded and later mixed with water to gain plasticity.

This method is mostly used for making sculptures.

Steps

i. Mining:

Clay is mined from its localities or deposits.

ii. Sorting:

Impurities such as stones, roots etc. are sorted from the clay.

iii. Drying:

Clay is placed in sunshine to dry so that its easily pounded.

iv. Wedging and Kneading:

Clay is then mixed, wedged and knead to plasticity; this stage involves turning and pressing clay to remove any air pockets and to make it a homogenous mass.

v. Storage:

The clay is then stored in polythene bags to conserve the moisture.

WET METHOD

In this method clay is dissolved in water to form slip or slurry.

Steps

i. Mining:

Clay is mined from its localities or deposits.

ii. Slaking:

This is the process of soaking clay in water for a period of 3 to 4 days, its slaked to reduce the lumps of clay into a homogenous mass.

iii. Sieving:

After slaking, the clay is mixed with a wooden stick into slip; it is then sieved to remove any impurities. The finely sieved clay is then put in a bucket and at this point grog or kaolin is added.

iv. Drying:

The slip is then left to dry to plasticity i.e. to partially dry to a malleable state. Dry pieces of plaster of paris can be added to absorb the excess moisture.

v. Wedging and Kneading:

The clay is then pressed and folded to remove any air spaces to plasticity.

vi. Storage:

It is then stored in polythene bags to prevent it from drying.

THE PLASTIC METHOD

It is common in the brick and pottery industry. Here, moisture available in clay is utilized and if the clay is too dry, water is added to it. Mixed clay is directly used for forming products by hand throwing, coiling etc.

THE SEMI DRY METHOD

It combines dry and plastic methods. Here, clay is crushed to powder and then grog is added to it. Clay is then mixed with water till plasticity is attained. This produces a homogenous mixture.

PHYSICAL STAGES OF CLAY

I. Green ware:

This refers to dry and unfired clay articles

II. Leather hard:

It refers to clay that has been dried partially, at this stage the clay body is approximately 15% of moisture

III. Bone dry:

It refers to clay bodies which are completely dry with 0% moisture content

IV. Bisque ware:

It refers to clay after it has been fired into a kiln for the first time.

V. Terracotta

This is reddish brown half backed clay.

VI. Slip

This is clay in a liquid form.

ACTIVITY 22: *List the importances of clay artworks to the;*

- *economy of Uganda*
- *the community*

- *artist*

ACTIVITY 23: *List the importances of clay as a material of art and design.*

ACTIVITY 24: *List the places where clay can be mined from?*

CERAMICS AND POTTERY

Pottery refers to the clay artworks that are fired at low temperatures below 1000c.

Ceramics refers to the clay artworks that are fired at high temperatures that are above 1200c.

TOOLS USED MAKING CERAMICS AND POTTERY

- Potter's wheel
- Buckets for keeping clay
- Sieves
- Modeling tools for smoothening and decorating
- Rolling pin for making slabs
- Wedging table, to wedge and knead clay.

ACTIVITY 25: *Draw illustrations of the above tools as used in ceramics and pottery.*

METHOD OF PRODUCING CERAMICS AND POTTERY

Pinch method

This is when a lump of clay is rolled into a ball and then pinched using the thumb and fingers to form the shape.

Steps

- a) Wedge and knead clay thoroughly until all the air is expelled

- b) Take a lump of wedged clay and roll it into a ball
- c) Hold the ball in the left hand and push the right thumb into the middle of ball to make a hollow.
- d) Gently pinch and pull the walls of the ball while turning it around to form the required shape making sure that the walls are kept even.
- e) After forming the required shape, let the work dry partially in a cool dry place
- f) When the article is leather hard i.e. partially dry and no longer pliable, it can be decorated.
- g) Allow the article to partially dry before firing.

Coil method

The coil technique involves the use of rolled up clay which is coiled and built successively upon each other to create the required shape.

Process

- a) Wedge and knead the clay to plasticity
- b) Roll the clay on a flat surface into even sized lengthy coil pieces, equivalent to the size of a pencil.
- c) Score the base with a sharp instrument and add slip to it.
- d) Add coils to the base and press the inner sides of the coils to merge them to the base
- e) Add more coils to the article as you form the walls of your pots. Seal all the gaps and smoothen the article as you build it.
- f) When the required shape and height is obtained, smoothen the outer surface with a smooth tool, preferably the back of a spoon.
- g) Dry the article to leather hard and decorate as required.
- h) Allow the article to dry before firing
- i) Glaze after the first firing.

SLAB METHOD

This technique is used to make pottery from rolled clay slab cut into required shapes. Angular, circular shapes can be obtained which are joined together at all sides using slip to make the required shape.

Process

- a) Wedge and knead the clay to plasticity
- b) Prepare a flat surface to work from and cover it with a polythene material
- c) Get a small lump of clay and flattened it in your hands then place it on the flat surface and make it smooth and even by using of a rolling pin.
- d) Using a ruler and a knife, cut the slabs into required shapes and sizes
- e) Score the sides of the slabs and join the together using slip. Smoothen them into each other at all corners, both inside and outside of the form.
- f) When required shape and size have been obtained, allow the article to dry to leather hard and decorate
- g) Let it dry thoroughly before firing
- h) Glaze after the first firing.

THROWING METHOD

This involves the use of a potter's wheel on which clay is placed and an article formed while the wheel is rotating. There are two types of wheels; the kick wheel and the electric wheel.

Process

- a) Wedge and knead the clay, and make it slightly stiffer than that used in hand building techniques.
- b) Throw the clay to the center of the wheel. If the clay is not centered properly at the beginning, the forms will be difficult to manipulate and will be distorted.
- c) Rotate the wheel by kicking or switching on power.
- d) Dip the hands in water and cup the hands over the clay and press down at the same time. Water acts as a lubricant while working on the form.
- e) Pull up and shape the clay cylinder to the desired shape while rotating the wheel
- f) Smoothen the pot by wetting a sponge and move it up and down as the article rotates.
- g) Stop the wheel, use a wire to cut to release the pot from the wheel when the desired form has been made
- h) Let the pot dry to lather hard and decorate as desired
- i) Let it dry thoroughly in a cool dry place before firing
- j) Glaze after the first firing.

CASTING METHOD

This is a technique of producing ceramics by pouring slip into a mould that gives the required shape.

HOW TO CAST POTTERY

- i. Select a theme and a source of inspiration
- ii. Make a sketch and Design the ceramic form.
- iii. Collect and prepare the materials and tools to be used.
- iv. Make the mould which is going to be used for casting from plaster of paris because it has the ability of absorbing water quickly from clay.
- v. Prepare slip by mixing finely ground clay with water to a light cream consistency
- vi. Fit the pieces of moulds together and pour slip into the channels of the moulds and allow the moulds to stand for a period of time.
- vii. The moulds will absorb the excess moisture from the slip and a thin layer of clay will form on the inner walls of the moulds and will take on their shape. As the clay dries in the mould, it shrinks in size and detaches itself from the plaster mould
- viii. And when the form is dry, remove it from the mould to expose the finished article.
- ix. Decorate as required, glaze and fire to finish.

ACTIVITY 25: *With the above explanations, come up with relevant simple illustrations of each of the above method of making pottery or ceramics.*

DECORATING CERAMIC OR POTTERY

Inlaying

Ceramics forms can be inlaid with cut out shapes which can be of the same colour as the clay body. Use slip to fix these inlaid shapes to the clay article

Incising

Shapes can be cut into the clay form using suitable instruments.

Sgraffito and engobe.

The sgraffito technique is a form of scratching a design on the surface of a ceramic form. Any suitable tool can be used such as a pin, nail

Painting: this can be very elaborate. The paint is often applied to pottery that has been fired once, and may then be overlaid with glaze afterwards.

Burnishing

When the form is leather hard, it can be polished by rubbing it with any smooth surface. Burnishing enriches the color of the article.

Impression

Design can be created by impressing objects into moist clay forms. Patterned shapes can be used e.g. wood blocks once the impression has been made, colored slip can be brushed into the groove.

Slip trailing

Different colored clay slips can be slip-trailed on the surface of the ceramic forms. They can be applied by dipping, spraying, sponging or painting.

GLAZING CERAMIC ARTICLES

Glaze is a vitreous substance used to cover ceramic wares in order to make them attractive, durable and impervious to liquids

TYPES OF GLAZES

Transparent glaze

This type of glaze is clear and the colour of the clay body or under glaze is seen.

Opaque glaze

This type of glaze is non-transparent and it obscures the fine details of the forms

Gloss glaze.

Gloss glazes are smooth and glass like; they are highly reflective and brilliant.

Matt glaze

They are porous and rough glazes

FIRING

This is the process of baking ceramic articles with heat in a kiln in order to harden it or fix an applied substance such as glaze. During firing the clay changes from green ware to bisque ware i.e. changes from a raw material to a permanent material.

TYPES OF FIRING

Bisque fire:

This is the first firing of raw ceramic ware. When fired, the chemicals and water in the clay molecules is drawn out and the clay fused together. At this point, clay will not disintegrate again to become plastic.

Glaze firing:

This is a type of firing where ceramic articles are first glazed before firing. During this firing, the glaze is transformed into a coating of glass which fuses to the clay. Wax is applied to the bottom of the glazed forms to prevent them from sticking to the kiln rack because the glaze melts during firing.

HOW TO FIRE CERAMIC AND POTTERY

Firing pottery is done in a kiln. It can be done using a variety of methods, with a kiln being the usual firing method. Before firing pottery, make sure that it dries completely. Pottery can be fired in the following ways;

USING A PIT FIRE KILN

The pit fire kiln is used in the following ways;

- To fire pottery in any kiln, the art work should be totally dry.
- Start by digging a large pit so that it's able to house all the pottery art works.
- Go ahead and fill the bottom of the pit with saw dust.
- Place the pottery each facing in a different direction from the other.
- Pile up different layers of saw dust around each pottery art work.
- Light a fire from the top of the pit and allow it to burn slowly till they attain the bisque form.
- Once burnt, allow the pottery to cool then dig them up.
- Clean up the pottery so that it's ready to use.

USING A BORNE FIRE KILN

The bonfire kiln is used in the following ways;

- This type of kiln is totally different from the other kilns.
- There is no need of digging pits for this kiln.
- Once the pottery is dry, arrange it in a heap.

- Pile up layers of dry grass on a leveled ground.
- Arrange the fire wood around and at the top of the heap of the pottery.
- Light up the fire to burn the pottery till it turns to a bisque form.
- Once burnt, allow the pottery to cool.
- Clean up the pottery so that it's ready to use.

USING AN ELECTRIC KILN

The electric kiln is used in the following ways;

- To fire pottery in any kiln, pottery art work should be totally dry.
- The electric kiln must be switched off from the electric currents or power supply.
- Arrange the different pottery into the kiln starting with the smaller sizes to the bigger sizes.
- The kiln should be able to close after arranging in the pottery.
- Once it's closed, switch on the electricity so that the kiln begins the firing process.
- Gradually check with the levels of heat within the kiln so that the pottery is not over heated as it may lead to major breakage.
- Once burnt, allow the pottery to cool before it's got out.
- Clean up the pottery so that it's ready to use.
- One can go ahead and glaze the pottery repeating step 2 to 7.

ACTIVITY 26: *Research and find the importance of each the following when it comes to ceramics and pottery;*

- | | |
|--------|-----------|
| • Grog | • Slaking |
| • Slip | • Scoring |

ACTIVITY 27: *Research and find out the causes of breakages in ceramics and pottery.*

SCULPTURE

This is the art of shaping representative/abstract forms in the round or in relief especially by modeling clay, chiseling stones, curving wood, cutting metal or plaster.

TOOLS AND MATERIALS USED IN SCULPTURE

NOTE: A tool is an instrument or device that is used in the making but not part the final artwork.

A material is a substance from which the artwork is made from.

The tools and materials used for sculpture include;

- | | | |
|---------------|--------------------|---------------------|
| • Clay | • Sponge | • Sand paper |
| • Wood | • Ice | • Grinders |
| • Paper | • Plastics | • Axle blade |
| • Cement | • Paints | • Gloves |
| • Plaster | • Saw dust | • Over-roll / Apron |
| • Metal | • Wax | • Gauges |
| • Wires | • Mallet or Hammer | • Spade |
| • Fiber glass | • Panga | • Chisels |
| • Soap | • Bow/ Hand saw | |
| • Stones | • Any drawing tool | |

TYPES OF SCULPTURE

There are two types of sculpture and these include;

- Relief sculpture
- Sculpture in round

RELIEF SCULPTURE.

This is a type of sculpture that is curved in or out of the surface.

Types of relief sculpture

i. **HIGH RELIEF (ALTO – RELIEF)**

This is a type of relief sculpture where the artwork is almost completely carved from its surface.

ii. **LOW RELIEF (BAS – RELIEF)**

This is a type of relief sculpture where the artwork is slightly projected out of the surface. The artwork barely extends past the base e.g. images on a coin.

iii. **SUNKEN RELIEF (INTAGLIO)**

This is a type of relief sculpture where the artwork is carved into a surface so that its highest projection is below the surface.

SCULPTURE IN THE ROUND.

This is a type of sculpture that exists independently in a complete three-dimensional form and not attached to a support or a background, it can be seen from all corners.

ELEMENTS OF SCULPTURE

1) Solids.

These are three dimensional masses which occupy spaces, they are areas occupied by the main body

2) Voids.

These are spaces which are partially or fully enclosed by planes. Voids are the negative spaces through a sculpture or depression while solids are positive masses.

3) Planes.

Planes refer to the area of a surface which is defined by a more or less abrupt change in direction; a plane can be flat or curved. In sculpture several flat and curved planes are organized in different directions to create a unified form.

4) Contours.

These are the outlines of objects in spaces. They are also junctions of planes.

PRINCIPALS OF DESIGN USED IN SCULPTURE

Principals of sculpture are just as important as in other areas of art and design, these include

- Balance
- Dominance
- Repetition

- Proportionality
- Movement
- Contrast
- Harmony

Balance

Balance that makes sculptural form stable. A sculpture must be made stable to stand without toppling over when placed on or stands on the ground.

Movement

Movement refers to the rhythm of the sculptural form and can be along vertical, diagonal, or horizontal planes. This enables the sculptor to see more interesting forms with space.

Contrast

Greater contrast is achieved by playing one principal against the other in different proportions. Contrast is not achieved by having equal proportions of elements.

Unity and harmony.

Unity, harmony refers to the overall relationship of elements and principals which gives a three or two dimensional form stability and visual interest.

Proportionality

It refers to relationship of parts of a sculpture as compared to another or its self.

TERMS USED IN SCULPTURE

Armature: It is an internal frame work designed to support the material used in making the body of sculpture

Marquette: A Marquette is a small scale model or rough draft of a sculpture to represent a bigger project of sculpture.

Pedestal: Is the base on which a freestanding sculpture is mounted. It can also be built as part of the sculpture from below.

Base: the portion of the work on which the sculpture rests.

Chisel: a sculptural tool used in the subtractive process. Chisels are used by sculptors for carving stone, wood, and other materials.

Freestanding: a sculpture intended to be viewed from all sides.

Hand tool: is a device for doing a particular job that does not use a motor, but is powered solely by the person using it.

Life size: refers to art work or sculpture made with same size of a person or object represented.

Mold: a negative (hollow) form that is filled with a material such as slip, plaster or metal and removed when the material hardens into the shape of the mold. A mold can be used to make copies of an object.

Raku: it's a clay firing technique where sculptures are fired at high temperatures then removed to cool when red hot, resulting into a cracked glaze effect.

Replica: An exact copy or reproduction of an artwork, sometimes made on a smaller scale than the original.

Statue: a carved, modeled, or sculpted three-dimensional figure, especially of a person or animal, that stands up by itself.

METHODS USED FOR MAKING SCULPTURE

There are two methods used for the making sculptures and these include;

- **ADDITIVE METHOD**
- **SUBTRACTIVE METHOD**

➤ ADDITIVE METHOD

This is a method that involves the addition of materials to build up a form to the required composition. There are two main techniques of the additive method;

- i. **MODELING:**
This is the process of building up form from malleable materials such as clay, plaster of paris etc.
- ii. **CONSTRUCTION/ ASSEMBLING:**

This is the process of building up form by assembling or fitting together materials. These materials can be joined together by welding, nails, glue, stitching etc. the materials which may be used are; wood, metals, stones, plastics, gourds, leather, fabric etc

➤ **SUBTRACTIVE METHOD**

This method involves cutting away of parts of a mass of material until the required shape (form) is created. The following are the techniques of the subtractive method;

a) CASTING

This is a method of producing three dimensional forms by pouring liquid or molten materials into a mould that gives the required shape.

A **mould** is a device used for shaping material in molten or plastic state into the required form.

TYPES OF MOULDS

i. Press mould.

This is a mould that shapes materials in plastic state.

ii. Slip mould.

This requires the use of slip or any molten material.

How to cast forms using the slip mould using clay

- i. Select a theme and a source of inspiration
- ii. Make a sketch and Design the sculptural form.
- iii. Collect and prepare the materials and tools to be used.
- iv. Make the mould which is going to be used for casting from plaster of paris because it has the ability of absorbing water quickly from clay.
- v. Prepare slip by mixing finely ground clay with water to a light cream consistency
- vi. Fit the pieces of moulds together and pour slip into the channels of the moulds and allow the moulds to stand for a period of time.
- vii. The moulds will absorb the excess moisture from the slip and a thin layer of clay will form on the inner walls of the moulds and will take on their shape. As the clay dries in the mould, it shrinks in size and detaches itself from the plaster mould
- viii. And when the form is dry, remove it from the mould to expose the finished sculpture.

- ix. Decorate as required to finish

b) CARVING

This is a subtractive technique of producing sculpture by cutting or chipping away parts of the material until the required forms are made. This technique is mostly applied with wood carving.

WOOD CARVING

This is a subtractive method where wood is cut and chipped away until a required form is achieved.

MATERIALS USED FOR WOOD CARVING

- | | |
|---------------|----------------------|
| - Wood | - Paint |
| - Wax | - Wood preservatives |
| - Vanish | - Wood glue |

TOOLS USED FOR WOOD CARVING

- **Machete/ panga**
- **Chisel:** It is used to cut and remove parts of wood by placing a sharp edge against the wood and pushing or pounding the wood end with a mallet.
- **Gauge:** Gauges are round blade wood carving tools. Used to carve or cut round shaped edges.
- **Coping saw:** is used to cut off chunks of wood at once.
- **Clamp:** is a wood carving tool with a clasp. Used for holding and strengthening wood piece being carved in one position.
- **Mallet:** small wooden hammer with a big round or box head at one end and a short handle used for driving the carving tools i.e. the chisel, gouge into the wood etc.
- **Sandpaper:** A strong sheet of paper coated with sand or other abrasive material for smoothing and polishing the surface of wood sculpture especially in the difficult to reach areas.

- **Hand axe:** Hand held tool consisting of a head flattened to a blade that has one or two blades and short handle for chopping wood and roughing out.

PREPARATION OF WOOD AS A MATERIAL

- Cut the wood into sizeable pieces, remove its bark when still wet and place it in a dry place.
- Coat the edges with mineral oil or melted paraffin to prevent development of cracks.
- Loosely wrap the wood in old cloth pieces in order for it to lose moisture. Let the wood season for about a month.
- Place the wrapped wood in shelter, let it season for one more month.
- Unwrap and expose it in shelter for 2-3 days
- Carve the wood into any product that you want.
- Varnish to create a good finishing for the art work.

PHASES IN WOOD CARVING

The process of carving goes through number of stages. These include;

The Inspiration stage

This stage involves conceiving the idea for carving i.e. the object to be carved. Sketches are developed in numbers until the final is achieved.

Preparation stage

Prepare the log for carving i.e. by drying it first and sealing off the ends to avoid 1cheeks and cracks to develop during the process of carving. Prepare the necessary tools to used and get them ready for use.

Roughing out stage

This stage involves preliminary carving, removing the buck of the log if necessary, and making simple outlines of the different planes and proportions with grooves all around the outline of the object.

Intermediate carving

Cut away the mass of wood around the desired form leaving it raised up while the curved portion gets sunken. Continue dealing with detailed studies.

Final finishing

Finishing is done by using a rasp file, sand papering, waxing and painting if necessary.

ACTIVITY 28: *Research and find out the advantages and limitations of the following materials as used for sculpture;*

- wax
- Cement
- Soap
- Sponge
- Paper
- Metal
- Clay

ADVANTAGES OF USING WOOD FOR SCULPTURE

- Can be permanent if well treated.
- It can withstand areas of tension in a given artwork
- Being a natural material, it can be easily accessed.
- Wood grains may add to the texture of the finished work.

LIMITATIONS/ DISADVANTAGES OF USING WOOD FOR SCULPTURE

- Wood requires number of tools and other materials to be carved into sculpture which makes it expensive
- If not well treated it can develop checks and cracks during the process of carving
- Wood does not permit corrections unless a change in design. Any cut piece is irreplaceable.

RELEVANCY / ADVANTAGES OF SCULPTURE

Sculpture has been made for various purposes for generations in societies in Africa and other parts of the world. The following are some of the roles of sculpture;

- 1) Sculpture expresses ideas, feelings and experiences of an artist. It enables him to bring out his deep feelings and emotions.

- 2) Sculpture also communicates what an individual aspires to reach e.g. sculpture made about ideal beauty of form that individuals aspire to reach.
- 3) Sculpture plays a role of reflecting man's activities in the society. It reflects its economy, politics, religion, culture, plants and animals.
- 4) It's monumental. This type of sculpture is made to commemorate an event or someone great, such sculpture is usually big in size and significantly seen in the area where it is placed.
- 5) Sculpture plays an important role in conserving the culture of people. It helps to conserve history, religion, customs and economic activities of people.
- 6) Sculpture gives a special type of beauty in the place where it's located.
- 7) Sculpture can also be used for academics, simplified to suit the artist's interpretation of the elements and principles of art.

FABRIC DECORATION

Fabric decoration is the patterning an essentially plain fabric to render it more appealing to serve a particular purpose. This includes any of the many concepts and activities used for enriching a fabric.

METHODS OF FABRIC DECORATION

- Printing
- Tie and dye
- Batik
- Embroidery
- Appliqué
- Patch work

REASONS FOR DECORATING FABRICS

- Fabrics are decorated to make a plain piece of fabric look interesting
- Fabrics are decorated to differentiate social classes
- Fabrics are decorated according to the culture that has been put in them.

People base on historical and ancestral believes.

- Fabrics are also decorated to suite the social classes of people and also recognize companies, institutions and organizations from the way they dress and present themselves.
- Fabrics are also decorated to earn a living.
- Fabrics are also decorated for communication purposes. They are decorated especially for advertisements by different organizations and individuals

TEXTILE PRINTING

This is a surface resist process of reproduction of texts and images with ink on paper, cloth or any other materials. The process is carried out in four basic techniques namely;

- **Relief method:** This is printing made by a raised surface
- **Intaglio:** This is printing made from sunken areas

- **Lithography:** This is printing made from a flat surface
- **Serigraphy:** This is printing made through a surface

TYPES OF PRINTING

There are three methods of textile printing and these include;

- Screen printing
- Stencil printing
- Block printing

MOTIF

A motif is a repeated pattern used in a design during printing.

Procedure for motif development

- Select a theme or a source of inspiration for the design. This can be got from nature or artificial settings of still life or daily human activities
- Draw the inspiration on paper and study and formulate other relevant sketches.
- Simplify the drawing into outlines.
- Introduce positives and negatives by shading black for the positives to make the motif ready for printing



ACTIVITY 29: Research and find out the meaning of the following terms as used in textile printing;

- *Half drop*
- *Full drop*
- *Repeats*

BLOCK PRINTING

This is a method of textile printing where a wooden block or linoleum-block bearing a design is used. The block is subsequently coated by thickened paste which is transferred or applied to the surface of a fabric. It's a technique that works also in text printing, images and patterns.

Materials used for Block printing

- Fabric
- Paste
- Thickeners
- Printing ink

Tools used for Block printing

- Carving tools i.e. chisel
- Rubber gloves
- Flat iron
- Rollers
- Printing blocks
- Wooden stamp
- Printing table
- Cutter

Processes of Block printing

- i) Select a theme and source of inspiration for your design
- ii) Develop a motif, and identify the negatives and positives
- iii) Trace the design on the wooden or linoleum block and using any drawing tool.
- iv) Using cutter or any other cutting tool, carve out the negative parts of the design so the positive parts stand out.
- v) Prepare the fabric and pin it to the printing table.
- vi) To print, apply printing paste to the block by rolling out printing material onto a flat palette.
- vii) Then transfer the printing material to the surface of your block and press it firmly and steadily on the cloth.
- viii) For flow of pattern and repeats, ensure perfect registration of the motif.
- ix) After printing leave it to dry before ironing.

ACTIVITY 30: *Research and find out the advantages and limitations of using block printing.*

STENCIL AND SCREEN PRINTING

STENCIL PRINTING

Stencil printing is a method of fabric design where designs are cut in a stencil and (printed) transferred to the fabric.

A stencil is a sheet that cut with design or patterns used for textile printing. Some of the improvised stencil are cardboards, manila papers, photographic films, x-ray films etc.

Tools used for Stencil printing

- stencil
- Razor blades and cutters
- Sponge
- Squeegee: this is a rubber tool that is used to press printing ink through a screen to a prepared surface.
- Masking tape
- Flat iron

Materials used for Stencil printing

- Cloth or fabric
- Printing paste
- Water
- Thickeners
- containers

Process or procedure for stencil printing

- a) Select a theme and source of inspiration.
- b) Develop a motif, and identify the negatives and positives
- c) Place the paper with your motif on a flat surface table and attach/ staple your stencil
- d) Cut out the positives on the stencil using a cutter or a razor blade.
- e) Place the cut stencil on top of the fabric to be printed
- f) Add printing paste on a sponge and press through the cut gaps
- g) Repeat the motif throughout the cloth to cover it with the design
- h) When done, dry the cloth in a cool dry place before ironing.

ACTIVITY 31: *Research and find out the advantages and limitations of using stencil printing.*

SCREEN PRINTING

It is the art of using a mesh and a screen to transfer ink/paste on to a surface using a squeegee.

Tools used for Screen printing

- Screen
- Squeegee
- Printing table/ light table
- Container
- Sponge

Materials used for Screen printing

- Cloth
- Printing paste
- Water
- Thickeners

Procedure for making a printing screen

- Select a theme and develop a motif, and identify the positives and negatives.
- Prepare a stretched mesh on a wooden frame to form an empty screen
- Spread a thin layer of photo emulsion on the screen and let it to dry in a dark place for about 45mins
- Scan the motif to a computer and print it to a transparent material for example tracing paper.
- Put the screen upside down on top of a black surface, then put your transparency with positives and cover it with a glass block
- Expose the screen to light, the light causes the emulsion to harden and bind to the screen. Where the light passes makes a solid layer and where the light is blocked by the positives, remains water soluble.
- Wash and spray the screen with water only in the areas of the positives; this clears the areas where the ink will be pressed during printing.

Procedure for screen printing

- Select a source of inspiration/ theme
- Develop a motif and identify the positives and negatives
- Transfer your motif to the screen with photo coat/ photo emulsion.
- Prepare the materials and tools to be used.
- Lay a cloth on a printing table
- Place the screen at the starting point of printing and apply just enough printing paste on a squeegee and make the first print
- Continue the printing throughout the cloth
- When complete, Wash the screen after printing and leave the cloth to dry and iron it after

ACTIVITY 32: *Research and find out the advantages and limitations of using screen printing.*

TIE AND DYE

Tie-dye is a bound resist method of decorating a fabric. In the process, a piece of fabric or cloth is tied and dyed typically using bright colors. The fabric is tied, folded or stitched to prevent penetration of dye.

Materials used for Tie and Dye

- Cloth or Fabric
- Dyes or dylons
- Nylon threads
- Salt
- Water
- Dye bath

Tools used for Tie and Dye

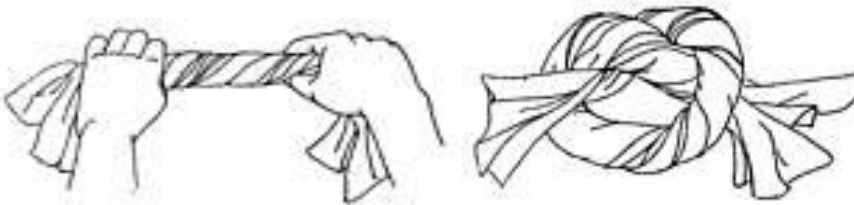
- Heat source
- Flat iron
- Objects such as stones and bottles tops
- Container

- Needles
- Nylon threads and rubber bands or raffia
- Gloves
- Face masks

Techniques used for Tie and Dye

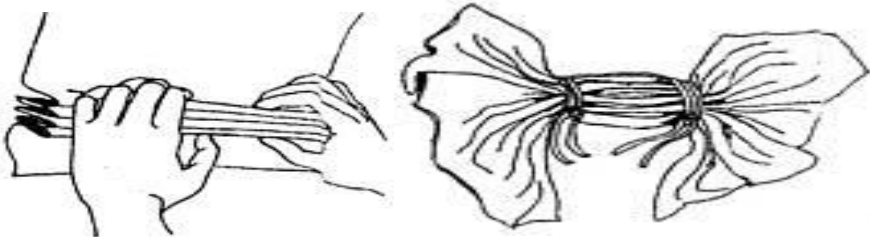
i) Knot tying

Hold the cloth at both ends and twist into a long rope form, tie this long rope into a knot and tighten as much as you can without damaging the cloth. You can tie as many knots as you have room for. Rubber bands or string can be tied over the knots to reinforce them as well as provide fine lines in the pattern.



ii) Pleats or Pleating

Lay the cloth on a flat surface and fold it into small folds (pleats) this is done following the length of the cloth or diagonally from one corner of the cloth, be careful not to lose any pleats and then tie accordingly. Loop rubber bands or string very tightly around all the pleats several times and knot.



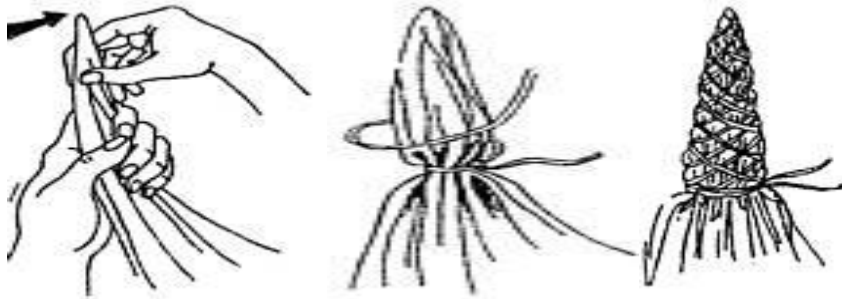
iii) Stripes

Lay the cloth on a flat surface, gather the cloth following its length and tie it



iv) Circles

Pull up a point on your cloth and twist it clock wise to form an anti-hill form. Tie it from the base up-wards. This can be done at several points on the cloth. The cloth can also be tied with objects of different sizes at different at different points.



v) Spirals

At the middle of the cloth, pinch a part and twist it to form a curl, and tie the entire cloth



vi) Stitchery

This is a technique of tie and dye where threads are sewed in the patterns drawn on the fabric and tightened in order to resist the penetration of dyes.

Procedure followed for Stitchery

- Select a theme and source of inspiration
- Sketch your design out using a pencil on paper.
- Collect and prepare the materials and tools to be used.
- Transfer the sketch of your design out using a pencil on the fabric and create contours.
- Use a large needle and nylon threads stitch through the contours of the design by using running stitches.
- Tie a very thick knot at the end of the thread to prevent them from falling out.
- One stitch at a time, pull the fabric back towards the knot until it is bunched up.

- Pull the fabric as tight as possible. Secure the thread with another thick knot.
- Mix the dye according to the dye instructions
- Dip the fabric in the first colour of the dye.
- Dry in a cool dry place to prevent the dye from fading
- Pull other stitches in places you want to retain the first colour and dip in second colour of dye
- Dry the cloth and go through the same process until the design is got.
- Cut the stitches out to reveal your tie dye patterns.

ACTIVITY 33: *Using your practical experience with the tie and dye process, write down the procedures of each of the above techniques of tie and dye.*

ACTIVITY 34: *Using your practical experience with the tie and dye process, write down the advantages and limitations of using the above method.*

ACTIVITY 35: *Using your practical experience with the tie and dye process, write down the importances of tie and dye artworks to the community and to the economy of Uganda.*

BATIK

Batik is a "wax resist" process for making designs on fabric. After the wax dries, it prevents the dye from spreading to those areas of the fabric that have been waxed.

Materials used for Batik

- Wax
- Fabric
- Dye
- Water

Tools used for Batik

- Heat source
- A flat surface as a work area
- Papers
- Flat iron / iron box
- Pencils

- Tjanting tool
- Brushes different sizes
- Containers

Techniques used for Batik Making

1. OUTLINES

Hot wax is brushed on the fabric to fill solid shapes or to outline patterns. The lines can be varied in thickness and directions by controlling the flow or wax, a brush or tjanting tool is used.

Steps taken

- Select a theme and make sketches of the composition on paper.
- Collect the materials and tools to be used.
- Lay the cloth on flat surface and transfer your sketch on the fabric.
- Prepare the dye according to the dye instructions
- Melt the wax in a container
- Using a brush or tjanting tool apply wax on all areas you would like to maintain white and apply the first colour of dye to the entire cloth
- Spread the cloth to dry in a cool dry, to prevent wax from melting and the dye from fading.
- When the cloth is dry, apply more wax to places where you want to retain the first color of the dye and dip in the second colour of dye.
- Spread the cloth to dry in a cool dry, to prevent wax from melting and the dye from fading.
- When you are done with the design you want, Remove the wax from the cloth by squeezing and creasing the cloth
- Lay several layers of paper on a flat surface and the put the cloth on top. Cover the cloth with other papers and iron, keep on changing the paper until the excess wax is removed

2. CRACKLED AND MARBLED EFFECT

A fabric can be completely waxed then crackled and dyed to produce a cracked or marble effect.

Steps taken

- Select a theme and make sketches of the composition on paper.
- Collect the materials and tools to be used.

- c) Lay the cloth on flat surface and transfer your sketch on the fabric.
- d) Prepare the dye according to the dye instructions
- e) Melt the wax in a container
- f) Using a brush or tjanting tool apply wax on all areas you would like to maintain white and apply dye to the entire cloth
- g) Spread the cloth to dry in a cool dry, to prevent wax from melting and the dye from fading.
- h) When the cloth is dry, apply more wax to places where you want to retain the first color of dye, do this until the design is complete.
- i) When you are done with the design you want, apply wax to the whole cloth and the crackle the cloth when the cloth dries. Cracking is creating cracks in the wax on the cloth and paint the cloth with a dark color
- j) Dip the cloth in a dark colour preferably black and spread it out to dry.
- k) Remove the wax from the cloth by squeezing and creasing the cloth.
- l) Lay several layers of paper on a flat surface and the put the cloth on top. Cover the cloth with other papers and iron, keep on changing the paper until the excess wax is removed

3. TEXTURED EFFECT

This is the creating of texture effects in a batik artwork.

Steps taken

- a) Select a theme and make sketches for the composition on paper
- b) Collect and prepare the materials to be used.
- c) Prepare the dye according to the dye instructions.
- d) Lay the cloth on flat surface and transfer your sketch on the fabric
- e) Place the cloth on a rough surface and rub wax on areas you would like to maintain white and apply dye to the entire cloth.
- f) Spread the cloth to dry in a cool dry, to prevent wax from melting and the dye from fading.
- g) When the cloth is dry, rub more wax to places where you want to retain the first colour still on a rough surface.
- h) When you are done with the design you want, Remove the wax from the cloth by squeezing and creasing the cloth
- i) Lay several layers of paper on a flat surface and the put the cloth on top. Cover the cloth with other papers and iron, keep on changing the paper until the excess wax is removed.

4. SCRATCHED EFFECT

The fabric is waxed and then scratched with a blunt instrument to produce fine lines.

Steps taken

- a) Select a theme and make sketches for the composition on paper
- b) Collect and prepare the necessary materials and tools to be used.
- c) Lay the cloth on flat surface and transfer your sketch on the fabric.
- d) Mix the dye according to the dye instructions.
- e) Melt the wax in a container
- f) Using a brush or tjanting tool apply wax on all areas you would like to maintain white and apply the first colour of dye to the entire cloth
- g) Spread the cloth to dry in a cool dry, to prevent wax from melting and the dye from fading.
- h) When the cloth is dry, apply more wax to places where you want to retain the first color and dip in the second colour of dye.
- i) Apply the entire cloth with wax and scratch through the wax with a pin or any other blunt instrument and dip the cloth in a dark colour.
- j) When you are done with the design you want, Remove the wax from the cloth by squeezing and creasing the cloth
- k) Lay several layers of paper on a flat surface and the put the cloth on top. Cover the cloth with other papers and iron, keep on changing the paper until the excess wax is removed.

ACTIVITY 36: *Using your practical experience with the batik process, write down the advantages and limitations of using the above method.*

ACTIVITY 37: *Using your practical experience with the batik process, write down the importances of tie and dye artworks to the community and to the economy of Uganda.*

EMBROIDERY

This is a sewing technique used to decorate fabrics with a needle and a thread; it can either be achieved by machine or by hand

Tools used for Embroidery

- Needles
- A pair of scissors
- Thimble, a small cap which fits over the middle finger
- Frame

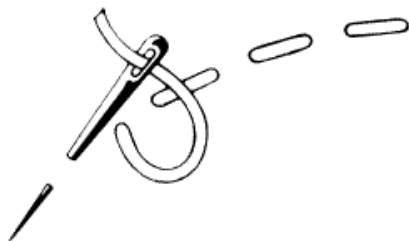
Materials used for Embroidery

- Threads of different colors
- Fabrics
- Raffia
- Sisal cotton
- Silk
- linen

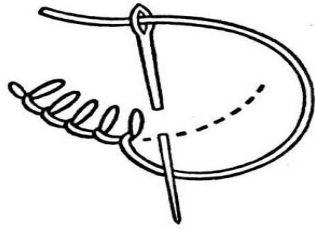
STITCHES USED FOR EMBROIDERY

In embroidery, a stitch is a single pass of a needle in a fabric when sewing. There are a number of stitches used in embroidery to give a fabric an interesting design depending on the purpose and type of design and these include;

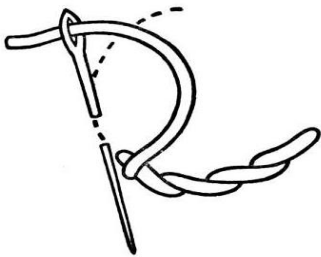
- **A running stitch:** this is worked by passing a needle in and out of the fabric. It is sometimes called a straight stitch.



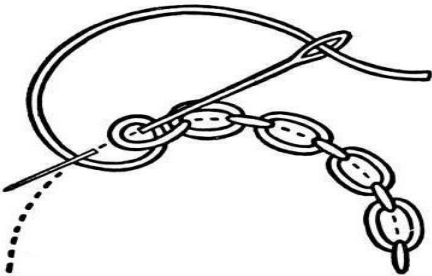
- **Blanket/button stitch:** here, one holds the rope of the thread on the surface of the fabric and the needle is returned to the back of the fabric at right angles to the original start of the thread.



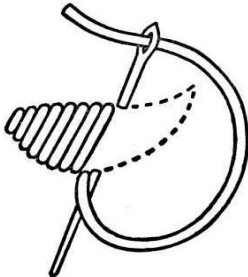
- **Cross stitch:** this is where a needle is inserted from the left to the right diagonally into the fabric so that slanting lines are formed. The cloth is formed by taking the thread and needle to cross in the previous stitch.
- **Stem stitch:** here, outlines of patterns are used to give raised effects.



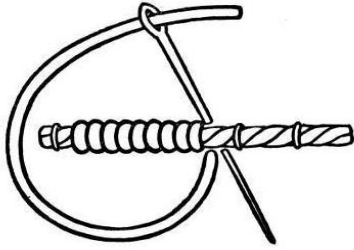
- **Chain stitch:** it's made by inserting a needle at a certain point in and out at another point and then reinserted at the same place to form a loop.



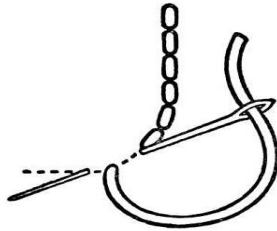
- **Satin stitch:** this is a flat stitch that is across a shape. These stitches lie closely together and in parallel lines.



- **Overcast stitch:** To work it, run down a thread on the traced line, then with fine thread cover this over with close upright stitches, picking up as little material as possible each time in order to make the line clear and round. The stitch is worked most perfectly in a frame.



- **Back stitch:** Back stitch sometimes makes a good line or outline. To work it, bring the needle through an inch from the end of the traced line, insert it at the start and bring it through again beyond where it first came out. Each stitch will be seen starting at the point where the last one finished.



Embroidery Techniques as used in fabric decoration

- **Flat embroidery:** patterns are drawn on the fabric and embroidered with basic stitches. Colours can be used to give illusions of flat and raised areas.
- **Quilting:** it's a three dimensional technique that consists of two layers of material caught together by lines of stitching. The padded fabric is embroidered to give a decorative effect.
- **Patchwork:** it's basically two dimensional. It involves the sewing together of small pieces of fabrics cut to define shape. The shapes are usually geometrically made so that the joining is easy. They are sewn end to end.
- **Drawn thread work:** it's another embroidery technique. It involves drawing threads from a piece of material. The areas where the threads have been removed are embroidered to give interesting effects

Procedure of making an Embroidery Artwork

- Select a theme and a source of inspiration.
- Make sketches of your composition on paper and suggest the colors to follow.
- Collect the necessary materials and tools to be used.
- Transfer your sketch to the fabric to be embroidered.

- Using relevant stitches, embroid the fabric following the sketch.
- When complete, trim off the unwanted threads to give an interesting finish.

APPLIQUE AND PATCH WORK

Patchwork is the sewing together of small pieces of fabrics cut to define shape onto a larger design. The shapes are usually geometrically made so that the joining is easy. They are sewn end to end.

Applique refers to the method of fabric decoration in which another piece of fabric is sewn, stapled, fixed to another fabric to create patterns.

Materials and Tools used for Patchwork and Applique

- | | |
|------------------------------|------------|
| • Cloths (colored and plain) | • Threads |
| • Wool | • Ruler |
| • Paper | • Needles |
| • Sisal | • Scissors |
| • Skins and hides | • Pencil |
| • Bark-cloth | • Thimble |

Procedure used for Patchwork

- Select a theme and source of inspiration
- Make sketches of your composition on paper.
- Collect and prepare the necessary materials and tools to be used.
- Cut out different designs and materials of colored fabrics and hem the hedges neatly making sure that each piece equals the rest
- Join the pieces together using any stitch worked on the wrong side
- When complete, trim off the unwanted threads and fabrics to give an interesting finish

Procedure used for Applique

- Select a theme or source of inspiration for your applique
- Make sketches of your composition on paper
- Collect and prepare the materials and tools to be used.
- Cut out different colored fabrics according to your design and sketch

- The colored fabrics are then sewn onto the larger fabric/ ground fabric
- When complete, trim off the unwanted parts to give an interesting finish.

Techniques of Applique

There are several methods of appliqué:

- **Reverse/ inverse applique:** this is a method of applique where a design is cut out of a larger fabric and a different fabric is sewn below the gap.
- **Direct applique:** This is where a pieces of a fabric are sewn directly on top of the larger fabric.
- **Machine applique:** here, a machine is used to sew the fabric pieces.

Procedure used for Applique

- Select a theme or source of inspiration for your applique
- Make sketches of your composition on paper
- Collect and prepare the materials and tools to be used.
- Cut out different colored fabrics according to your design and sketch
- The colored fabrics are then sewn onto the larger fabric/ ground fabric
- When complete, trim off the unwanted parts to give an interesting finish.

ACTIVITY 38: *Using your practical experience with applique, write down the advantages and limitations of using it.*

ACTIVITY 39: *Using your practical experience with applique, write down the importances of applique artworks to the community and to the economy of Uganda.*

WEAVING

Weaving is the process of producing a fabric by interlacing a set of vertical threads (warps) with a set of horizontal threads (wefts) on a loom.

Tools Used for Weaving

- Loom: this is a hand or machine operated device for weaving.
- Shuttle: it's a tool used to interlace the weft threads through the warp threads.
- Beater: it is used to press the weft threads together.
- Cardboards
- Pair of scissors

- Sticks
- Nails

Materials Used for Weaving

- Threads
- Raffia
- Sisal
- Wool
- Raphia
- Banana fibers etc

BASIC TERMS USED IN WEAVING

- **Warps:** these are threads that run vertically on a loom.
- **Wefts:** these are threads that run horizontally on a loom.
- **A pick:** This is one weft thread.
- **An end:** this is one warp thread.
- **Draft:** this is a plan made for the weaving process.
- **Shedding:** this is the process of separating warp threads to create space for the pick to run through.
- **Shed:** this is a gap (space) between the warp threads. The ends are separated by raising or lowering threads to form a clear space where the pick can pass
- **Web:** a piece of fabric decorated by weaving.

IMPORTANCE OF WEAVING

- It's a source of income and employment
- It expresses the feelings of an artist
- Weaving preserves culture and tradition
- It brings out someone's creativity.
- It's used in making and decorating fabrics craft work helps us to occupy leisure time with a beneficial activity.
- For domestic functions like wearing and mats for sitting on.

LOOMS

A loom is a hand or machine operated device for weaving.

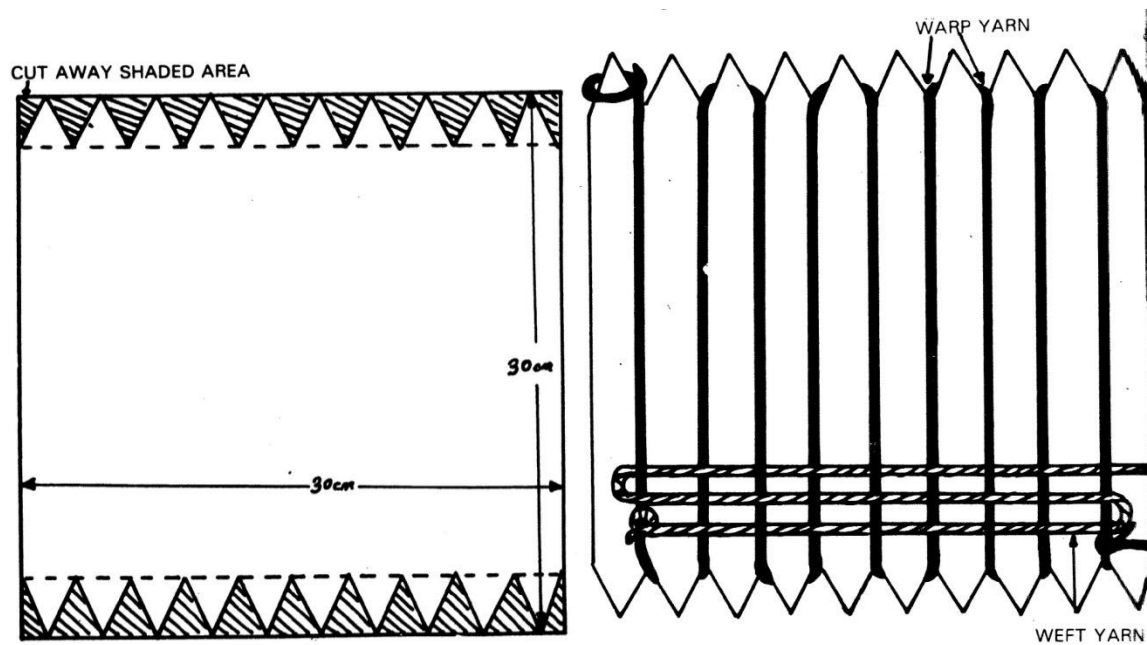
TYPES OF LOOMS

- Paper loom
- Hand loom
- Table loom
- Card loom
- Serrated loom
- Frame loom
- Dowel loom
- Inkle loom
- Heddle loom

Making a simple weave using a serrated loom

The basic tools and materials are strings of natural fiber such as sisal, pencil, a card board paper and a cutter.

- Create equally spaced markings of 1cm at the top and bottom sides of a cardboard box in equal dimensions and sizes
- Cut open each marked area and create gaps in which warp yarn shall be fitted during weaving. The attained cuts may be triangular or just straight from a single cut.
- Pick a yarn and then roll or tie it well on one corner of the first narrow opening. Fit it up and down in each cut slit on both ends of a card board and fill it with warp yarn.
- Use a needle or a similar object to fit in weft yarn and carefully start to weave by interlacing the weft yarn through the warps, side by side using a needle in a continuously way.
- When complete, fasten or cut off all loose ends of the thread to finish the weave



TYPES OF WEAVES

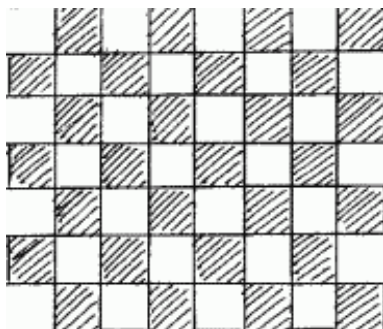
a. Plain weave

Also known as **tabby or taffeta weave** is basically one that involves one weft thread going under one warp thread and under another warp thread in succession across the width of the cloth.

Types of plain weaves

i) Balanced weave

This is a type of plain weave where warps and wefts show equally



ii) Warp faced weave

This is a type of plain weave where there are so many more warp threads than weft threads that the weft is all but covered by the warp.

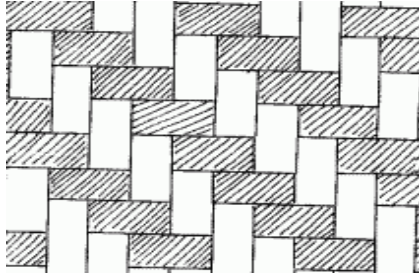
iii) Weft faced weave

This is a type of a plain weave where the warp is all but covered by the weft. To accomplish this, space the warps far enough apart that the rows of weft will pack down and cover them.

b. Twill weaves

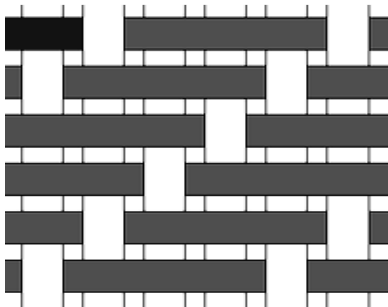
Twill weave structures create diagonal lines of pattern in the cloth. These types of weaves create floats of wefts over a number of warp threads. There are two basic types of twill weaves and these include;

- i) **2/2 twill, also known as a balanced** weave because both the right and wrong side of the cloth appear identical. It's when one weft thread runs over two warp threads, under two, over two, under two across the width of the fabric



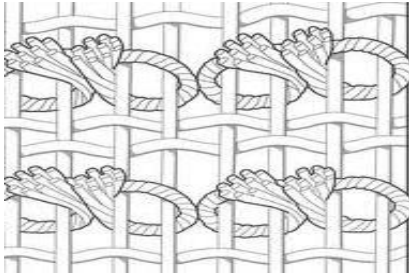
- ii) **3/1 or 1/3 twill. Also referred to as an unbalanced.**

It's when one weft thread runs over three warp threads, under one, over three, under one across the width of the fabric.



c. Ghiordes knot

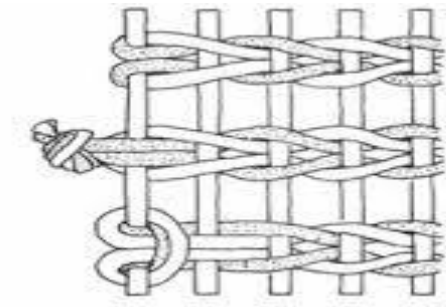
A knot is tied around two adjacent warp threads, and each row of knots is followed by a row of plain weave which holds them in place.



d. Twining weave

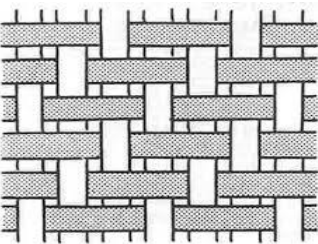
This technique is where a figure of eight is formed around the warp threads every time they intertwine with two weft threads. The twining weaves gives a very dense cloth, so it's suitable for rugs and carpets.

The twining technique is used to weave baskets and rugs.



e. Satin weave

In a satin weave, one warp yarn is floated over four or more weft yarns tied down with one thread resulting in a smooth face.

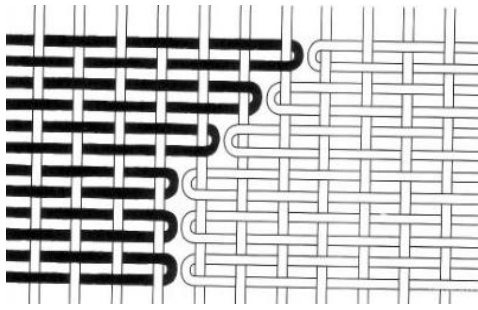


f. Tapestry weave

This is a weft-faced weaving technique where all the warp threads are usually widely spaced and hidden in the completed work. There are various types of tapestry techniques some of which are discussed below;

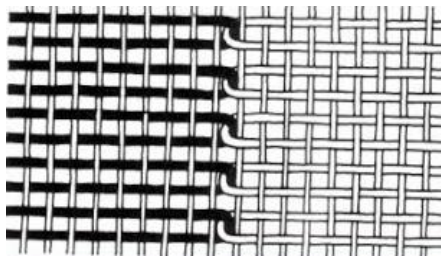
i) Slit tapestry weave

Here, slits are left before sections of warps usually in block of colours. Used to create regular shapes and straight patterns.



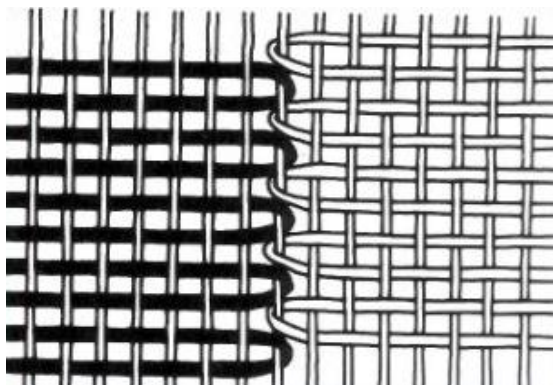
ii) Interlocking tapestry weave

in this technique, two different coloured wefts/textured wefts are used together to form patterns which interlock where they meet. They create irregular shapes.



iii) Dovetail tapestry weave

here, wefts that adjust one another are joined and are brought around to adjacent warps. It's mostly used in weaving irregular shapes.



ACTIVITY 40: Using your practical experience with weaving, write down the advantages and limitations of the above method.

ACTIVITY 41: *Using the explanation about weaves, create a plain and a twill weave using a paper loom.*

ACTIVITY 42: *Using the explanation about looms, research, draw name the parts of a frame loom.*

PAPIER-MÂICHÉ

This is a material consisting of paper pieces or pulp, mixed with glue.

Materials used for papier maiche

- Paper
- Glue
- Water
- Paint

Tools used for papier maiche

- Wires
- Saucepan for boiling and soaking papers
- Mortar and pestle
- Sieve
- Paper cutters
- Paint brushes etc

PREPARATION METHODS

Two main methods are used to prepare papier-mâché;

1. LAYERED/ STRIP METHOD

This method makes use of paper strips glued together with adhesive.

Procedure for The Layered/ Strip method

- Select a theme and a source of inspiration
- Make a sketch of the composition
- Create mould
- Prepare the desired paste/ glue
- Tear papers into strips
- Cover the mould with polythene papers to stop the glue or adhesive from sticking to it,

- Dip one strip at a time into prepared paste and stick it over the mould.
- Completely cover the form with a layer of strips overlapping each other.
- Layer the paper strips in alternating directions, to make the structure stronger.
- After a layer or two are applied, let them dry for 24 hours before more layers are added to get the desired layer effects. This avoids your shape buckling as it dries
- Paint and decorate finished form as required.

2. PAPER PULP METHOD

This is where paper is soaked or boiled with glue so that it softens.

HOW TO MAKE PAPER PULP

- Soak torn up or ripped up paper in a container of water overnight.
- When the paper softens, crush it into paper pulp using a blender or mortar. Only crush up a cup or so of paper at a time, and make sure there's plenty of water.
- Strain the mixture through a sieve. Gently squeeze out the extra water, but not too much. The pulp should be firm and damp, not hard and dry.
- Mix the pulp with glue, use cassava flour or wood glue.
- Apply the paper mache pulp over the prepared armature or mould.

PURPOSES OF PAPER MAICHE

- For making costumes, masks for festivals and other occasions
- Making toys
- Making props for schools and theatrical plays
- For decorations purposes

ADVANTAGES OF PAPER MAICHE

- It's a cheap material since it can be prepared from wasted paper
- It's a light material

LIMITATIONS OF PAPER MAICHE

- It requires a long preparation process in order to be used
- It cannot survive in moist environment
- Requires reinforcement for big pieces and areas of tension
- Can only be used indoors

MASKS

- These are crafts made to imitate faces of different creatures.
- It's also a craft normally worn on the face, typically for protection, disguise, performance, or entertainment.

METHODS OF MAKING MASKS

Mask making has various methods depending on the purpose which include;

- Wood carving
- Casting
- Papier maiche

Uses of masks

- decoration i.e. wall hangings
- entertainment i.e. drama, films, etc
- disguise to hide personality, character, mood and identity
- spiritual /worshiped as gods

ACTIVITY 43: *Using your knowledge about sculpture, come up with the procedure followed when a mask using;*

- *wood carving*
- *casting*
- *clay*

ACTIVITY 44: *Using the explanation, list any six types of masks.*

ACTIVITY 45: *Using the explanation, list the products made from papier maiche.*

PUPPETRY

Puppetry is an act where dolls and manipulated objects are used to make fun through imitations and movements.

Materials and Tools used for making puppets

- empty boxes
- empty cans
- threads
- wood
- hammer
- Hand saw, if the material is wood
- Pair of pliers
- Pair of scissors
- Painting brushes
- Wood
- Small nails
- Binding wires
- Leather
- Nylons or cotton strings
- Colored cloth
- Paints
- Socks

TYPES OF PUPPETS

The most common types of puppets are string puppets, rod puppets, hand puppets, shadow puppets, and ventriloquists' figures

i) MARIONETTES/ STRING PUPPETS

These are puppets operated by means of strings attached to their hands, legs, head and the body from a control, usually a wooden device held above the stage by the puppeteer.

ii) Rod puppets

These are controlled by rods attached to their limbs, heads, and bodies.



iii) Hand puppets

These are worn like a glove over a puppeteer's hand and arm. Much of a hand puppet's movement is controlled by the fingers and wrist of the operator.

iv) shadow puppet

it's a cut-out figure held between a source of light and a translucent screen. Shadow puppets can form solid silhouettes or be decorated with various amounts of cut-out details.



Carnival or body puppet

These are puppets designed to be part of large spectacles idealizing individuals, events. They are large and held by performers.

Procedure of making a marionette

Marionettes/ string puppets:

- Make sketches of the marionette to make and get the required materials form a piece of wire with loops on each end for the different body parts e.g. the thighs, legs, hands, arms, shoulders to act as joints for the marionette.
- Papier maiche the wires to make the head, torso, legs, thigh, upper arm, lower arm and the hand to bring out their form.
- Join the body parts together using a wire and loosely so that they can freely move
- Paint and give it any extra texture or effects to make the marionette ready
- Create a control. String up the marionette on the head and other body parts with a cross of wood to control.

IMPORTANCIES OR USES OF PUPPETS

- Used for entertainment especially in theater
- Used for advertisement and TV programs.
- Decoration especially in kids play centers
- Ceremonial or ritual celebrations like carnivals.
- In politics as dummies for contestants.
- They attract, capture attention to convey feelings and message.
- Used in teaching for dramatization or teaching aid especially in kindergartens.

- Religious and spiritual

BASKETRY

This is the art or practice of producing baskets by interweaving strips of pliable materials such as cane, palm leaf, sisal.

Materials and Tools used for Basketry

- | | |
|---------------|-----------------|
| • Sisal | • Scissors |
| • Papyrus | • Bamboo |
| • Grasses | • Cane |
| • Palm leaves | • Banana fibers |
| • Raffia | • needles |
| • Awl | |

Importances of basketry

Being a craft producing activity, it has the following importances;

- To earn a living
- For decorative functions
- For domestic functions like collecting food.
- Provides hands on practice
- They are used in entertainment
- Provides hands on practice
- They are used in entertainment craft work helps us to occupy leisure time with a beneficial activity.
- To preserve culture and tradition

Techniques used in basketry

Wicker work/ Splint technique: it involves the use of stiff materials such as cane or reeds to construct various forms such as trays, baskets etc. The materials are soaked in water before and used when still wet so that it's flexible and can be formed into a required shape. If used when dry, it will break.

Plaiting/ Checker technique: it involves the use of soft materials that are plaited and joined together to form baskets e.g. palm leaves, sisal raffia. Long strips are usually plaited or woven and later joined to form a mat or a basket.

Twinning technique: here, a figure of eight is used and two wefts cross each other at a warp thereby closing and interlacing it.

It involves the use materials like sisal that are twisted around warps threads. It involves the use of figure eight action in which two wefts threads cross each other at warp thread. The basket is twinned from the base upwards.

Coiling: here a basket is done by winding into a loop using grass, straws, sisal and rushes. This involves using soft materials like straws, reeds which are wrapped by another soft material which is sewn together to form a required shape. This method begins from the base and increases or decreases according to the size and shape required.

Factors considered when designing a basket

- **Function:** the purpose of the basket should be known.
- **Form:** a particular form of basket is determined to a large extent by its function with aesthetic functions considered.
- **Shape and size:** these determine the function and aesthetic references of the basket.
- **Decoration:** here the effects of the decorations are considered for a particular basket.
- **Technique:** this can influence the form of the basket. Some techniques are suitable for large projects e.g. in making a cane chairs.

Procedure of making an African coiled basket

- Collect some suitable grass or other materials and arrange them into a coil.
- Wrap the coil with Raphia, banana fiber or papyrus or other material to make a required coil.
- Form a base with the end of the coil by spiraling it and sewing into place by stitchery.
- Add more coils as needed and sew into place.
- Narrow / increase the circumference by using narrow or wider coils.
- Finish off the basket by using a neat over stitch.

ACTIVITY 46: *Using your knowledge about tools used in basketry, come up with relevant sketches of those tools.*

ACTIVITY 47: *Using the explanation about basketry, come with illustrations of the following;*

- A wickerwork basket
- A coiled basket
- A plaited basket
- A twined basket

ACTIVITY 48: Using the explanation about basketry, outline any 8 hand woven crafts.

ORNAMENTS

These are objects that are worn to enhance the appearance of a person.

Materials and Tools used for making ornaments

- | | |
|---|----------------------------------|
| • Wood | • Scissors |
| • Seeds | • Sharp knives |
| • Bones | • Awls |
| • Horns | • Needles |
| • Feathers | • Pliers for metal and wire work |
| • Shells | • Hammer |
| • Pieces of bamboo | • File for smoothening |
| • Reeds | • Tin snip for cutting metal |
| • Scrap iron, aluminum, copper, gold, brass | |

Importance of making ornaments

They are used;

- For social factors
- For economic factors like earning income
- For ritual purposes
- For cultural purposes
- For aesthetical purposes

Types of ornaments

These depend on the type of ornament being made and the material being used. They include;

- **Bead work:** they include knotted beads, strung beads, woven and stitched beads
- **Bone jewelry:** they include bangles. Here bones maybe cut and shaped into many forms.
- **Metal jewelry:** here copper, brass etc. can be cut and shaped, left plain or hammered to texture its surface.
- **Paper jewelry:** newspapers or gloss magazines pictures can be used to make beads and other forms of ornaments. They can later be painted and varnished.

Examples of ornaments

- Necklaces
- Anklets
- Bangles
- Earrings
- Bags and purses made from beads

Techniques of designing ornaments

The following are used when designing different ornaments;

- Painting
- Coiling using pliers
- Incising
- Drilling
- Casting

Factors considered when designing ornaments

- **Tools:** different ornaments require specific tools. These can cut, shape and alter the form needed for the ornament.
- **Style:** the purpose of the ornaments requires different styles. Styles can also affect the shape, colour, pattern of a specific ornament.
- **Function:** it's one of the major factors that are considered when making ornaments. One should know whether it's for social, political, ceremonial, ritual or that particular ornament.

- **Technique:** several methods can be used but they are also determined by the style, function and materials used.

Procedure of making a paper bead necklace

- With a source of inspiration, design the necklace and draw its sketch on paper with its approximate colours.
- Select the different papers to be used.
- Using a strong thread such as polyester and fishing lines, string the beads according to the design.
- Use a slip knot at one end of the thread to prevent the beads from slipping out.
- While the threading is complete, release the slip knot and join the two ends using a suitable knot.

LEATHER WORK

Leather is a material made from various types of animal skins, such as calf skin, goat skin sheep skins, and skins of wild animals.

Tools used in leather work

- Chalk
- Scissors
- Knife
- Awl
- Punching tools
- Mallet matting tools
- Brushes

PROCESS OF FORMING LEATHER

Preparatory stage: this where the hides and skins are prepared for tanning. It includes the following;

Skinning

The process starts with skinning the animal where the skin is removed from the slaughtered animal.

Washing

This involves soaking and washing the skin to remove dirt and impurities.

Dehairing and fleshing

The skin is softened by washing in lime to remove the hair and remain with the flesh.

Cleaning

The skin is subjected to another washing by scrubbing with a brush.

Tanning and curing

The skin is cured by subjecting it to a tanning process

Drying and compressing

This involves airing and drying the pelt, after which it's compressed under heavy rollers to give the leather firmness of texture.

Furnishing

It is the last stage which involves graining, where the surface is embossed with imitation, dyeing the leather and measuring to determine the size of leather.

Techniques of working with Leather**Dyeing**

Leather dyeing usually involves the use of spirit or alcohol-based dyes where alcohol quickly gets absorbed into moistened leather, carrying the pigment deep into the surface.

Painting

Here, the paint remains only on the surface. Due to this, leather painting techniques are generally not used on items that can or must bend nor on items that receive friction, such as belts and wallets because under these conditions, the paint is likely to crack.

Carving

Leather carving entails using metal implements to compress moistened leather in such a way as to give a three-dimensional appearance to a two-dimensional surface.

Stamping

Leather stamping involves the use of shaped implements (stamps) to create an imprint onto a leather surface, often by striking the stamps with a mallet.

Molding/Shaping

Leather shaping or molding consists of soaking a piece of leather in hot or room temperature water and then shaping it by hand or with the use of objects or even moulds as forms. As the leather dries it stiffens and holds its shape.

Perforation

Perforation is the result of punching that is provided for the manufacture of a large number of regularly arranged openings of regular shape in the sheet and other material.

Modeling: this is a decorative technique made by raising parts on the surface leather.

Embossing: this is a decorative technique made by raising parts of leather from underneath.

Tooling: this is a decorative technique made by pressing lines on the surface of leather

TERMS USED IN LEATHER WORK

Pelt: The protective covering of an animal.

Hide: The pelt of large animals

Skin: Refers to the pelt of smaller animals.

Kip: this is the pelt (protective covering) of medium sized animals like heifers and bullocks.

Crust: Leather which has been tanned but not finished. Such leathers referred to as being in the crust.

Drum Dying: The application of dye stuffs to leather by the immersion. This process allows full dye penetration into the fiber.

Raw Hide: A hide which has only been treated to preserve it prior to tanning.

Liming: This process includes removal of the hair, preparing the hides for the tanning process.

Qualities of Good leather

Good leather should have the following qualities;

- It should have a few spots.
- It should be strong and durable.
- It should be flexible and should have a uniform colour.

- It should be attractive and relevant to what it is going to be used for.

Types of Leather

In general, leather is sold in these four forms:

- **Full-grain** leather refers to hides that have not been sanded, buffed, or snuffed (as opposed to top-grain or corrected leather) to remove imperfections.
- **Top-grain** leather is the second-highest quality. It has had the "split" layer separated away; making it thinner and more pliable than full-grain.
- **Corrected-grain** leather is any leather that has had an artificial grain applied to its surface.
- **Split** leather is leather created from the fibrous part of the hide left once the top-grain of the rawhide has been separated from the hide.

Procedure of making a leather product

A belt is band of leather or other materials worn around the waist.

Steps taken when making a leather belt include;

- Select the suitable leather for making the belt. Soft leathers are often preferable.
- Design the belt on paper. But consider sex, age and clothing on which it will be used.
- Lay the leather on a flat soft ground. Place the paper pattern on it and tape it.
- Using scissors or a knife cut the belt according to the pattern needed.
- Decorate the strip by any of the already practiced techniques.
- Cut holes along the middle line on one end of the belt.
- Fix the buckle on the other end by pushing the end through the center hole.
- Shape the other end of the belt to enter into the buckle for fastening.

GRAPHIC DESIGN

Graphics refers to the discipline of art that involves the designing of texts and images.

Terms used in graphics

- **Typography:** this is a technique of letter construction that involves designing, setting and arranging letters.
- **Calligraphy:** the art or practice of writing letters and words in decorative styles.
- **Book cover:** this is the outer layer of a book bearing information that summarizes the contents within that particular book.
- **Flaps:** these are continuations of the front and back cover that make up a book jacket.
- **Book jacket:** this is the outer most cover that is opted for instead of a book cover and it has a continuation of flaps on both the front and back cover.
- **Menu card:** this is a display card that shows items and services availed in an eating place like hotels and restaurants.
- **Wrapper:** this is an attractive cover of a product that is patterned with either texts or images mostly used when presenting gifts.
- **Poster:** it's a form of advertisement often displayed on walls or any flat surface to show information that should attract attention and convince viewers about something.
- **Calendar:** it's a form of art work advertising services through illustrations, letters and numbers. It's mostly used by companies to advertise their services and products.
- **Logo:** it's a symbol used by a company to identify themselves. It may be in words or illustration formats.
- **Trademark:** it's a name or symbol that identifies a particular product officially registered and restricted to the owner. It may be in words or illustration formats. It identifies products of a company from another company.
- **Monogram:** it's an identifying sign or mark of a person or a company that is designed following its initials. The letters used are often intertwined or twisted.
- **Emblem:** it's a symmetrical design consisting of visual symbols and mottos of slogans used to identify associations, trade unions, government parastatals etc.
- **Badge:** it's a symmetrical design or identification symbol made to be worn to show identification of a particular group like a school, club or institution.
- **Illustration:** it's a visual image used in graphics that helps to explain a specific theme. These form the pictorial part of the graphical work.

- **Packaging Design:** this is a design made for storing and transportation of goods and products.
- **Cartoons:** these are drawings used as illustrations in graphics that express humor and emotions that capture the essence of a particular statement.
- **Lay-out/format:** it's the general arrangement, plan or design of any graphical art work.
- **Motif:** it's a single design of a repeated pattern. It can be used when designing a wrapper.
- **Composition:** it's the combination of principles and elements of art by a visual artist to organize an art work.
- **Subject:** these are the main points of a graphical art work as chosen by the visual artist.
- **Content:** it's the message that the visual artist wants to communicate.
- **Target group:** the audience that the visual artist is designing the work for.
- **Concept:** it's the basic idea of any graphical art work.
- **Colour scheme:** it's the choice of colours used by the visual artist.
- **Emphasis:** it's a principle of art used by visual artist to make a certain part of an artwork to stand out from the rest of the work.
- **Focal point:** it's that part of the graphical work that is intended to grab the viewer's eye.
- **Positive space:** it's the space covered by the art work.
- **Negative space:** space without any art work.
- **Pattern:** a result got after a motif is repeated.

***ACTIVITY 49:** Using your experience, explain the tools and materials used in graphics.*

***ACTIVITY 50:** Using your experience, come up with a scrapbook.*

Types of posters

A poster is a form of advertisement often displayed on walls or any flat surface to show information that should attract attention and convince viewers about something. The following are the different types of posters used;

- **Informative posters:** these tell the audience about something they don't know about.
- **Formative posters:** these spread awareness about an issue mostly a business or a new brand or product.
- **Digital posters:** these are not printed but are projected or displayed screens for the public to view.

- **Campaign posters:** these are used to show information and awareness about ongoing campaign. Like cancer and hunger campaigns.
- **Fashion posters:** these generally say a lot about fashion products and brands.
- **Political posters:** these are used for political awareness mostly for elections.

Characteristics of a good graphical artwork

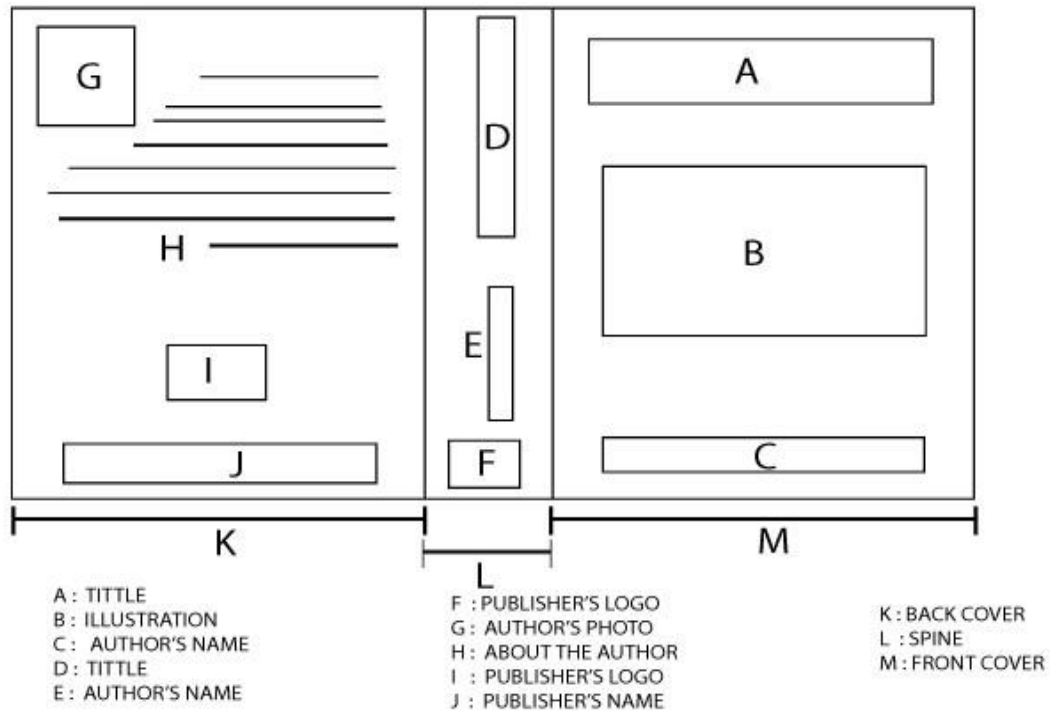
The following are the characteristics of a good graphical art work;

- Any graphical art work has to be bold, clear and visible. Take an example of a billboard that if it wasn't visible, road users could hardly read the displayed information. This is done through the choice of fonts and illustrations used.
- Graphical art works must be simple. These works shouldn't be complicated. The visual artist has to consider viewers of all ages even the kids in that they also have to understand the message depicted.
- Any graphical art work has to have enough space. It should have balanced illustrations or images that must fit with the design.
- Graphical art works must be attractive, appealing, representative and symbolic. They should arouse the interests of those that must view it so that it carries their attention. This is done mostly with the colour pallet choice.
- Creativity is the other trait of a good graphical art work. Let the art work be occupied with new ideas that are not borrowed from another work.
- Be original in the source of ideas and design. Do not copy from the already existing designs.
- Easy to recognize and remember.
You should use fonts and colour schemes which are easy to appreciate and remember.
- Meaning and Idea: Make the visual symbol easy to remember and use in different forms.

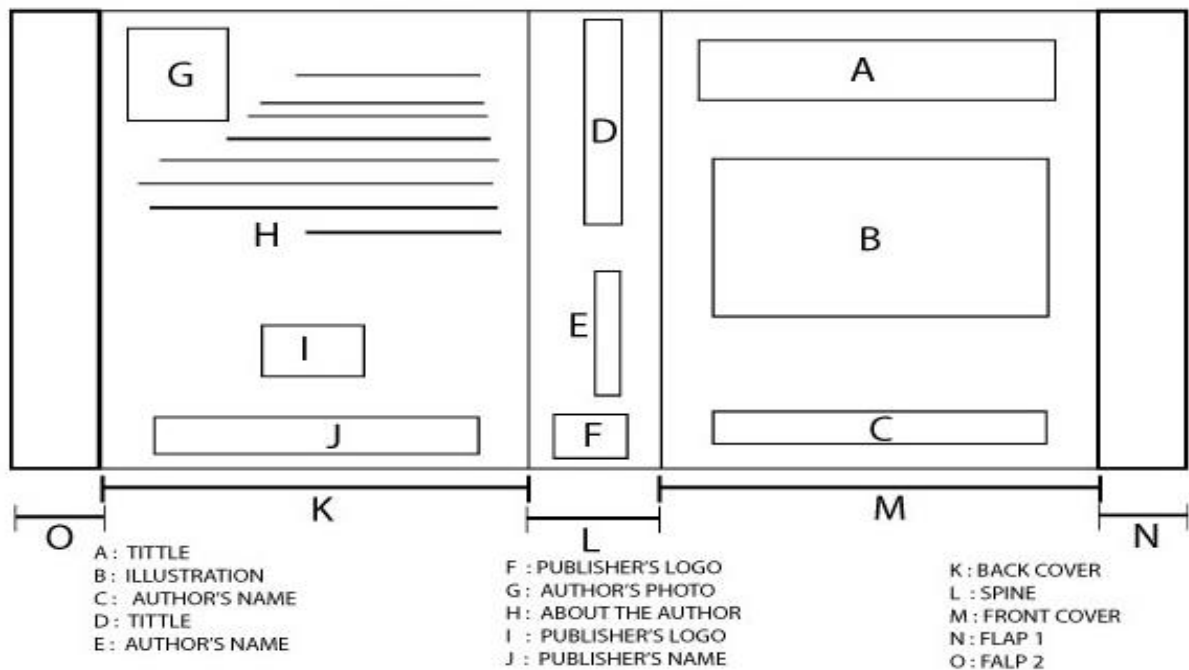
Characteristics of a good logo and trademark

- Both logos and trademarks should be bold enough in order to be easily identified.
- They should be simple and precisely designed.
- They should be unique with a touch of originality.
- Both logos and trademarks should be attractive with the initials of the companies or institutions they represent.

BOOK COVER FORMAT



BOOK JACKET FORMAT



UNEB PAST PAPERS

612/1 (UNEB 2020) STUDIO TECHNOLOGY

Define the following term as used in Art and design;

- | | |
|-------------|---------------|
| a. Contrast | d. Unity |
| b. Pattern | e. Proportion |
| c. Variety | |

Explain the following methods as used in sculpture:

- | | |
|----------------|------------|
| a. Additive | c. Casting |
| b. Subtractive | |

Define the term weaving?

Explain the importance of weaving to your community.

With illustrations, describe the process of preparing clay for pottery.

Explain five ways of improving the quality of art and crafts in schools.

What is meant by the term “resist process” as used in batik production?

Mention four similarities between batik and tie n dye.

With the help of diagrams, show the differences between an updraft and a downdraft kiln.

Why should a kiln be fully loaded before firing?

What precautions should be taken before loading a kiln and during the firing of green ware art pieces in a kiln.

What is a logo?

Mention four qualities of a good logo?

Define the term “screen printing”

Briefly explain the process of photo emulsion on a silk frame.

Using illustrations, name and describe how many three tools used in carving.

Give three limitations and three possibilities of using Plaster of Paris for carving.

P615/4 (UNEB 2019) STUDIO TECHNOLOGY

Define the term basketry?

List five materials and five tools used in basketry?

Mention five uses of baskets in your community?

With the help of illustrations, describe four types of chisels used in wood carving.

State any three disadvantages of using stone in carving.

Outline six different types of paper use in graphic art.

Using illustrations differentiate between two dimensional and three dimensional art works.

Describe four techniques used in firing clay wares.

Explain six steps that a finished ceramic ware undergoes to ensure even drying.

Define the following terms:

- Slip
- Green ware
- Terracotta
- In laying
- Plane
- Void
- Line
- Solid
- Contour

What is a craft?

Outline six materials and six tools used in the making of crafts.

Give the advantages of using local materials in the production of crafts in Uganda today.

What is the meaning of the following properties of colour?

- Hue
- Intensity
- Value

612/1 (UNEB 2019) STUDIO TECHNOLOGY

Define the term a studio as used in Art and Crafts?

Outline three aspects of conduct you need to observe while in the Art room.

Give five reasons why it is important to fire clay works?

Explain four post firing techniques of decorating pottery articles?

Using your knowledge of colour, list the sets of colour in the;

- i. Primary category
- ii. Secondary category
- iii. Tertiary category

Differentiate between organic shapes and geometrical shapes.

Define the term ornament?

List four materials and four tools used in making ornaments.

Using illustrations, explain what is meant by;

- a) Plain weave
- b) Twill weave

Explain four techniques used in tie and dye?

Give four materials used in the production of tie and dye?

Describe the process of applique production

Give four tools used in appliques.

What is meant by the term embroidery as used in crafts?

List three tools and three materials used in embroidery?

Explain five benefits of craft shops in your community?

Suggest five ways of promoting the craft industry in Uganda.

P615/4 (UNEB 2018) STUDIO TECHNOLOGY

List down five elements of art and design?

Give three uses of any one of the elements mentioned above.

What is a kiln?

Explain four types of kilns.

Differentiate between:

- i. Relief sculpture and sculpture in round
- ii. Additive and Subtractive methods of making a sculpture.

Why should sculpture made in be fired?

Compare the production process of Mosaics to that of Appliques in terms of materials used.

Describe the methods used in the making of mosaics.

Explain the following terms used in fabric decoration:

- i. Screen printing
- ii. Stencil printing
- iii. Patterns

Using illustrations, explain the procedure of screen printing.
State seven factors that should be considered by an artist when making a desired craft work.
Draw an illustration of plain weave and explain how it works.
Give four examples of leather work.
Describe the process of making a leather bag.
List down two materials and tools used in making leather work.
What is basketry?
Explain four techniques of making baskets.
Define the term Papier Mache
Explain how Papier Mache is made.

612/1 (UNEB 2018) STUDIO TECHNOLOGY

Define the term texture?
With illustrations, explain two types of textures?
What is meant by the term Batik?
List four materials and four tools used in the production of batik articles?
Explain the following methods as used in pottery;
 i. Pinch
 ii. Coil
 iii. Slab
What is meant by the term tesserae?
Give four similarities between mosaic and collage?
Explain five advantages of using wood as a sculpture material?
Name and draw four tools as used in wood carving?
Define the following terms as used in weaving;
 i. Yarn
 ii. Weft
Mention four materials and four tools used in weaving?
Describe the term crumpling as used in tie and dye?
Give four limitations of tie and dye as a method of fabric decoration?
Describe the processes taken in preparing clay for pottery?
Describe the following types of puppets;
 i. Body puppets
 ii. Sock puppets
 iii. Shadow puppets

Several factors hinder the development of art and craft in Uganda. With relevant examples, explain six factors that affect the development of Art and crafts in your community?

P615/4 (UNEB 2017) STUDIO TECHNOLOGY

Define a line as used in art and crafts?

With illustrations, discuss six importances of using lines in the making of crafts?

Mention four types of clay?

Discuss four properties of clay?

List four advantages of using wood as a material in sculptural production?

Explain four functions of sculpture in your community?

What is colour?

Draw or paint a colour wheel and label the following:

- i. Primary colours
- ii. Secondary colours
- iii. Tertiary colours
- iv. complementary colours

Differentiate between designers Gouache and poster colour?

Explain the similarities between the two types of colours mentioned above?

Define a mask?

With examples state the functions of masks in your community?

What is an art room?

Explain six characteristics of a good art room?

List four materials and four tools used in tie and dye?

Describe the stages of making a tie and dye piece?

Give the artistic meaning of a sketch?

Explain four importances of making sketches in any design work?

Define textile printing?

Describe four methods used in textile printing?

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Define shape?

List down two categories of shapes used in design?

What does plasticity in clay mean?

What is meant by the term layout as used in graphic design?

State two wet and dry media used in drawing?

Explain how tertiary colours can be made?

What is the use of a squeegee in printing?

Describe the process of preparing clay for ceramics?

Mention three characteristics of good clay?

Define the term batik?

List down two materials and two tools necessary for making batik?

List down two types of sculpture?

Give two functions of sculpture?

List two media and two supports (ground) used in drawing?

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Explain any three properties of colour?

What is a sketch?

(b) Explain the importance of sketching for any work of art?

Describe the process of preparing clay for sculpture art work.

Define the following terms;

- Radial balance
- Asymmetrical balance
- Symmetrical balance
- Rhythm
- Dominance

Describe the process of making batik?

Define screen printing?

In an area measuring 6cm x 7cm, design a single motif derived from plants. This motif is to be used on a textile.

What are the advantages of clay in the making of sculpture?

Define the term weaving?

Define the term mosaic?

Explain the process of making a fibre mosaic.

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List six principles of art and design?

Name and illustrate three types of line.

State the functions of line in drawing.

Explain the differences between block printing and screen printing?

Briefly explain the process of making a textile design?

Describe three techniques in making ceramic objects?

Give the procedure of making either a woven basket or a coiled basket.

Mention three finishing techniques for making a weave?

Outline six materials used in basket making?

Define the following terms;

- Slip
- Design
- Stencil printing
- Unity

Give six functions of crafts in the society today?

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Explain the following principle of art and design;

- Balance
- Rhythm
- Harmony
- Proportions

Define the term collage?

Outline six stages of producing a mixed media collage?

What is papier maiche?

Explain five steps of producing a papier maiche art work?

Define the following terms in relation to clay;

- Kneading
- Slip
- Terracotta
- Green ware

Discuss three safety practices by an art student in the art room.

Describe the process of preparing wood for sculpture.

Draw and label three tools used in wood carving.

Define the term weaving

List four techniques of weaving.

Explain the following techniques used in tie n dye;

- Pleating
- Crumpling
- Circles
- Running stitch

Differentiate between the following terms;

- Relief sculpture and sculpture in the round
- Addictive and subtractive methods of making sculpture

What is the importance of firing sculpture made form clay?

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Define the term line as used in art and crafts.

State four functions of line to a visual artist.

List four different materials that are used in textile design?

Mention three materials used in tie n dye and three materials used in batik crafts.

Define the following;

- Squeegee
- Design
- Motif

What is clay?

Describe the techniques used in pottery production?

What is papier maiche?

Describe the process of making a mask?

With illustrations, write the meanings of the following terms as applied in weaving;

- Warp
- Weft
- Shed

- Draft
- Pick
- End

Write short notes on the following;

- Collage
- Mosaic
- Sculpture

Briefly explain two types of sculpture.

List four materials required for making a batik.

Mention five functions of crafts to your community.

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List six principles of art and design?

Discuss the importance of the principles of art and design in the production of an art work?

Define puppetry?

Mention any two materials and two tools used in producing puppets?

What is clay?

Explain the meaning of the following terms;

- Primary clay
- Secondary clay

List down three tools and materials used for wood carving.

Give two advantages of varnishing wood?

Differentiate between warp and weft yarns?

Describe the process of preparing macramé?

Explain the process of preparing clay for pottery?

Define the term weaving?

List down five techniques of weaving?

Describe the process of producing batik?

Using a well labelled illustration, distinguish between symmetrical and asymmetrical balance in 3-dimension sculpture?

Mention any six materials that are used in making of collage?